

Surprises, Shapes or Machines?

Forecasting the Market's Response to Macroeconomic Releases with Dynamic Time Warping (DTW)

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Abstract

Macroeconomic releases move futures prices almost instantly; is the news fully priced by then? Using intraday S&P 500, Nasdaq-100, Dow and 10-year Treasury futures, 2010–2026, we show that equity-index futures keep drifting in the direction of the consensus surprise after the initial jump. Trading this drift is profitable out of sample in all three equity indices after costs. Dynamic time warping (DTW) analogues of the price path do not predict the drift's direction, but their dispersion predicts its size. In Treasury futures the drift is below one tick. Entering at the release price mostly captures the untradeable initial jump.

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I Introduction

Scheduled macroeconomic releases are among the cleanest information events in financial markets. Their timing is announced months in advance, a consensus forecast is published before each release, and the news itself—the difference between the actual figure and the consensus—arrives at a known second. Prices of index and Treasury futures react within the first minute (Fleming and Remolona, 1999; Balduzzi et al., 2001; Andersen et al., 2003, 2007). The speed of that first reaction is well documented. Much less is known about what happens next: how much of the news is still unpriced once the first minute has passed, whether the remaining move is predictable from information that is public at that time, and whether it is large enough to be exploited once the cost of trading is paid.

These questions matter for two reasons. First, they measure the speed of price discovery in some of the most liquid markets in the world. If prices fully incorporated the release within a minute, any later move would be unrelated to the surprise. If they do not, the size and duration of the remaining drift tell us how long it takes markets to agree on what a number means. Second, the answer separates two kinds of apparent profitability. A strategy that trades on the surprise at the release price captures the first-minute jump, which no investor can trade; a strategy that enters one minute later captures only the drift. Backtests that do not distinguish the two can overstate what is achievable by a large factor.

We study these questions in four futures markets with different sensitivities to macro news: the E-mini S&P 500 (ES), the E-mini Nasdaq-100 (NQ), the E-mini Dow (YM) and the 10-year Treasury note (ZN). We use one-minute prices from July 2010 to June 2026 and the Bloomberg release calendar with consensus forecasts for 179 U.S. release lines. Our design has three features. First, every trade enters at the close of the first one-minute bar after the release, so the untradeable jump is excluded by construction. Second, every model parameter—which releases to trade, the sign of each release’s effect, regression slopes and pattern-matching thresholds—is re-estimated each year using only earlier data, and the last years of the sample are held out. Third, we compare four forecasting approaches that were developed independently by the authors on the same data, entry rule and cost grid: (i) the sign of the standardized consensus surprise; (ii) dynamic time warping (DTW) analogues of the price path just before entry, which ask whether similar paths in the past were followed by a rise or a fall; (iii) DTW “event games” built on longer paths around the release; and (iv) a machine-learning classifier. The comparison is informative because the approaches use very different information: the first uses only the published numbers, the second and third use only prices, and the fourth combines both.

Five findings emerge. First, the news is not fully priced after the first minute in equity-

index futures. After a hot CPI print (a headline surprise above half a standard deviation), ES falls about 37 basis points (bps) in the first minute and a further 15–18 bps over the next 30 minutes; NQ falls about 55 bps and then a further 20 bps, and YM about 31 bps and then a further 17 bps. Roughly a quarter of the CPI response arrives after the first minute. In ZN the post-minute continuation is only 4–5 bps.

Second, the consensus surprise predicts the direction of this drift out of sample. A strategy that trades the sign of the surprise of CPI alone, entering one minute after the release and holding for 30 minutes, earns a net Sharpe ratio of about 1.0 in ES and NQ over 2019–2026 after a cost of two ticks plus \$4.50 per round trip. Extending it to about ten release families selected each year from history raises the trade count tenfold, to about 100 trades a year per market, with Sharpe ratios of 0.65 (ES) and 0.71 (NQ). On the held-out years 2021–2026 at the lower cost assumption common among practitioners (a quarter tick per side) the Sharpe ratios are 1.09 and 0.75. The same code, applied without change to the E-mini Dow after these results were known, gives 1.06 on its broad book, and the three equity indices together give 1.07. For the CPI strategy, the direction of the surprise is worth about 13 percentage points of cumulative return relative to being long in exactly the same 30-minute windows.

Third, price-path analogues do not reliably forecast the direction of the drift. Their rank correlation with the next 30-minute return is between 0.01 and 0.04 in every market; the sign of their forecast was positive in 2018–2020 and negative in 2021–2026 in ES and NQ; and requiring the analogues to agree with the surprise lowers performance in the held-out years in ES and NQ and leaves it unchanged in YM. Run on YM with the same code, the analogue book loses even before costs. The dispersion of the analogues, by contrast, is a stable forecast of the size of the move.

Fourth, the drift in ZN is smaller than the cost of trading it. The surprise predicts ZN’s first-minute jump as well as it predicts ES’s (R^2 of about 0.30 at five minutes), and a gross edge exists after the first minute, but at about 2.7 bps per trade it is below the 3 bps round-trip cost of one tick. Tick size relative to the size of post-announcement moves acts as a limit to arbitrage.

Fifth, the entry assumption dominates reported performance. The same classifier positions earn a Sharpe ratio of 2.1 when entered at the pre-release price and 0.9 when entered one minute later; about 80% of the reported profit is the first-minute jump. Across approaches, daily returns are nearly uncorrelated, and the surprise and DTW books lose in different years, so an equal-risk combination beats the best single approach: 1.39 on ES and 1.35 on the three equity indices at a quarter tick, against 1.09, with half the volatility. At realistic costs the high-turnover pattern-based approaches reduce the combined performance,

and the equity portfolio keeps 0.85.

Related literature. Our paper contributes to three strands of research. The first studies the high-frequency response of asset prices to macroeconomic news. [Fleming and Remolona \(1999\)](#) and [Balduzzi et al. \(2001\)](#) show that Treasury prices adjust to announcements within minutes, with a burst of volume and volatility; [Andersen et al. \(2003, 2007\)](#) document that surprises move foreign exchange, bond and stock prices almost instantly and that the response depends on the state of the economy; [Gürkaynak et al. \(2005\)](#) show that monetary-policy statements move asset prices beyond the policy rate itself; and [Savor and Wilson \(2013\)](#) and [Lucca and Moench \(2015\)](#) document return premia on announcement days and before FOMC meetings. We focus on the part of the response that remains after the first minute, compare it across equity and Treasury futures, and ask whether it can be exploited net of costs. The second strand studies how announcement effects depend on market conditions ([Andersen et al., 2007](#)). We find strong in-sample state dependence of the CPI response in equity futures, concentrated in the untradeable first minute and detectable in real time only from 2022, and no out-of-sample gain from conditioning on it. The third strand concerns pattern-based forecasting and the evaluation of trading rules. Dynamic time warping ([Sakoe and Chiba, 1978](#); [Berndt and Clifford, 1994](#)) is widely used to match time series of different speeds; we provide a clean out-of-sample test of its directional content and separate it from its information about volatility. Following [Harvey et al. \(2016\)](#) and [Bailey and López de Prado \(2014\)](#), we treat the multiplicity of tested specifications explicitly, and we report random-sign and block-bootstrap inference ([Künsch, 1989](#)) alongside heteroskedasticity-robust regressions ([MacKinnon and White, 1985](#)) and false-discovery-rate control ([Benjamini and Hochberg, 1995](#)).

Roadmap. Section II describes the institutional setting and data. Section III documents price discovery after releases. Section IV sets out the evaluation protocol and the four forecasting approaches. Section V reports the results, Section VI interprets them, Section VII presents robustness checks, and Section VIII concludes. Appendix Appendix C collects additional figures and tables for each stage of the analysis.

II Institutional Setting and Data

A Scheduled releases and the consensus

The U.S. statistical agencies and private survey providers publish their main indicators on calendars fixed well in advance. Most price and labor-market data, including the Consumer Price Index (CPI) and the Employment Situation report (nonfarm payrolls, NFP), are released at 08:30 Eastern Time; survey-based indicators such as the ISM indices and consumer confidence are released at 10:00; and FOMC rate decisions at 14:00. Before each release, Bloomberg surveys professional forecasters and publishes the median forecast, which we use as the market consensus. Because the consensus is fixed before the release and the actual figure is published at a known second, the difference between the two is news to the market at the moment of the release.

B Futures data

We use one-minute OHLCV bars from Databento (CME Globex, `GLBX.MDP3`) for four contracts from July 2010 to June 2026: the E-mini S&P 500 (ES), E-mini Nasdaq-100 (NQ) and E-mini Dow (YM), rolled on the calendar, and the 10-year Treasury note future (ZN), rolled on volume. Table 1 summarizes the contract specifications. The markets differ in how they should react to the same news. YM, the Dow, holds fewer technology and more industrial and financial stocks than ES, and serves as a third equity market on which the ES results can be checked. A hot inflation print raises the expected path of the policy rate, which lowers the present value of future earnings—more so for the long-duration technology stocks that dominate NQ—and lowers Treasury prices. Strong payrolls raise expected growth, which is good for equities, but also expected rates, which is bad for bonds.

Table 1. Contract specifications and round-trip costs.

	ES	NQ	YM	ZN
Underlying	S&P 500	Nasdaq-100	Dow Jones Ind.	10-year note
Exchange	CME	CME	CBOT	CBOT
Tick size	0.25 points	0.25 points	1 point	1/64 point
Value of one tick	\$12.50	\$5.00	\$5.00	\$15.625
Round-trip cost, 2 ticks + \$4.50 (bps)	≈ 1.7	≈ 0.7	≈ 1.0	≈ 3.0
Buy-and-hold Sharpe ratio, 2019–2026	0.90	0.99	0.78	−0.23
Buy-and-hold maximum drawdown, 2019–2026	−29.5%	−38.3%	−32.2%	−23.6%

Notes: The round-trip cost converts two ticks of slippage (one per side) plus a \$4.50 commission into basis points of the contract value at the pre-release price, averaged over 2019–2026.

We keep sessions from 18:00 to 17:00 Eastern Time, dated by their close, and drop exchange holidays and sessions with too few bars. Minutes without a trade inside a kept session are filled with the last price. Kept sessions number 3,894 for ES, 3,797 for NQ, 3,754 for YM and 3,916 for ZN; together the four markets contribute about 21 million one-minute bars. Returns across a change of contract are set to zero, which removes the calendar-spread jump at each quarterly roll and leaves returns within a contract unchanged. A one-minute bar is labelled by its start: the bar labelled 08:30 covers 08:30:00–08:30:59 and closes at 08:31. Figure 1 shows the four markets over the sample, which spans the low-inflation years before 2020, the COVID crash, the 2022 inflation shock with the fastest monetary tightening in four decades, and the disinflation that followed.

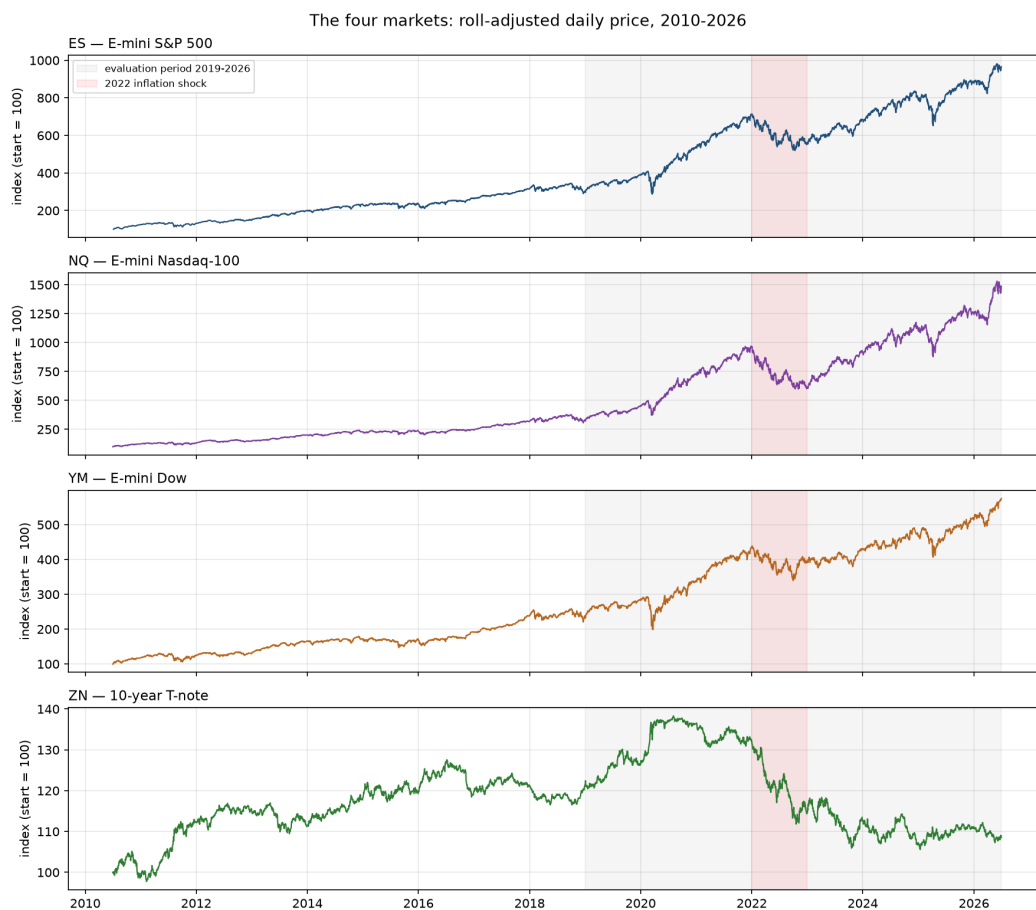


Figure 1. ES, NQ, YM and ZN, 2010–2026. Roll-adjusted daily prices rebased to 100. Grey: the 2019–2026 evaluation period; red: the 2022 inflation shock.

C Measuring surprises

The Bloomberg calendar contains 179 U.S. release lines and 27,137 release rows between 2010 and 2026, each with a release time, the actual value and the survey median. For line l at release t , the raw surprise $A_{l,t} - F_{l,t}$ is not comparable across lines (payrolls are measured in thousands of jobs, CPI in tenths of a percent), so we standardize it causally, using only earlier releases of the same line:

$$z_{l,t} = \frac{A_{l,t} - F_{l,t}}{\text{sd}(A_{l,1} - F_{l,1}, \dots, A_{l,t-1} - F_{l,t-1})}, \quad (1)$$

requiring at least 24 earlier releases and capping z at ± 5 (0.47% of the 17,328 standardized surprises are capped). Lines that are released at the same instant—for example the eight CPI lines—are grouped into a release family f when their release timestamps overlap by at least 80%. The family signal is

$$x_{f,t} = \frac{1}{|L_f|} \sum_{l \in L_f} s_l z_{l,t}, \quad (2)$$

where the sign $s_l \in \{-1, +1\}$ is the sign of the slope of a regression of the first-minute return on $z_{l,t}$, estimated separately for each market on earlier years only. With these signs, $x_{f,t} > 0$ always means good news for the market in question: a strong payrolls print is good news for ES and bad news for ZN.

D Sample

The event-study sample (Section III) contains 363 CPI, NFP and PCE releases from 2016 to 2026. The trading samples use all release families with at least 30 releases (75 families built from 143 lines). All strategies use the sample split in Table 2; the first three years only build the history that later estimates require.

Table 2. Sample split.

Period	Years	Use
Burn-in	2010–2012	analogue library and surprise history only
Train	2013–2017	first estimation window
Validation	2018–2020	out of sample (walk-forward)
Test	2021–2026	out of sample, reported once

III Price Discovery After Scheduled Releases

This section asks how much of the response to a release arrives after the first minute and whether the surprise predicts it. We define, for a release at minute T and horizon h , the reaction from the last pre-release price and the drift from the first post-release price,

$$R_{T,h} = \ln P_{T+h-1} - \ln P_{T-1}, \quad D_{T,h} = \ln P_{T+h-1} - \ln P_T, \quad (3)$$

where P_τ is the close of the bar labelled τ . The reaction is the sum of the first-minute jump $R_{T,1}$ and the drift, $R_{T,h} = R_{T,1} + D_{T,h}$. Only the drift can be traded.

A The jump and the drift

Figure 2 shows the average cumulative return around CPI releases from 2016 to 2026, separately for hot prints (headline surprise above $+0.5$ standard deviations, 38 releases) and cool prints (below -0.5 , 36 releases). Three facts stand out. First, all four markets jump in the direction theory predicts within the first minute: after hot prints ES falls about 37 bps, NQ about 55 bps, YM about 31 bps and ZN about 19 bps; after cool prints they rise by about 30, 42, 26 and 23 bps. Second, the equity indices keep moving in the same direction afterwards. After hot prints ES falls a further 15–18 bps, to about -55 bps 30 minutes after the release, and NQ a further 20 bps, to about -77 bps; YM falls a further 17 bps, to about -48 bps. After cool prints the continuation is smaller, about 3–6 bps in the three equity indices. Third, ZN continues by only 4–5 bps after hot prints and slightly reverses after cool prints. Before the release, all four markets are flat on average: there is no systematic pre-positioning in the 30 minutes before CPI.

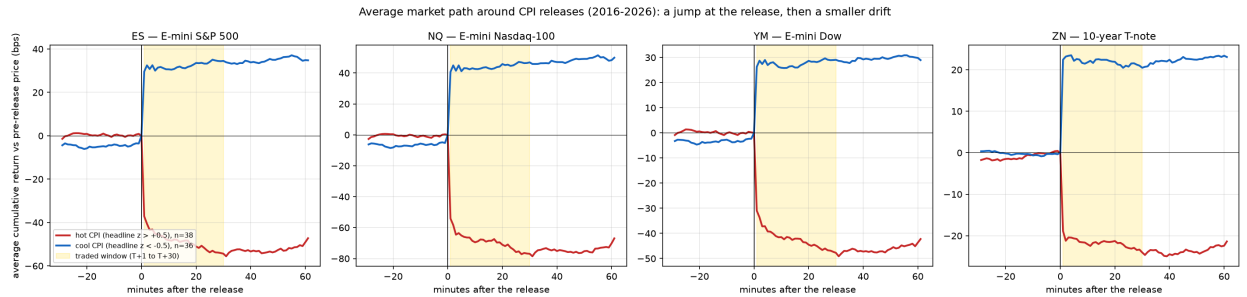


Figure 2. Average cumulative return around CPI releases, 2016–2026, relative to the last pre-release price, for hot prints (headline surprise above $+0.5$ SD) and cool prints (below -0.5 SD), ES, NQ, YM and ZN. Shaded: the window a strategy can trade (one to 30 minutes after the release).

B Surprises predict the reaction and the drift

For each release family we regress the reaction and the drift on the standardized surprise components with heteroskedasticity-robust (HC3) standard errors. Table 3 reports the results for CPI, NFP and PCE. CPI is the release that matters in all four markets: its headline surprise explains 27–30% of the five-minute reaction, with the expected sign. A one-standard-deviation hot surprise lowers ES by 20.8 bps, NQ by 27.7 bps and ZN by 9.9 bps over five minutes. NQ reacts about 35% more than ES, consistent with the higher rate sensitivity of long-duration technology stocks, and YM about 8% less (19.2 bps, R^2 0.27), with the same drift as ES after the first minute (−5.3 against −5.7 bps). NFP moves prices almost as much as CPI on average (30 bps in ES), but in a direction the surprise hardly explains (R^2 of 1–3%), because payrolls, the unemployment rate and wages often point in different directions.

Table 3. Response to CPI, NFP and PCE surprises, 2016–2026.

Market	Release	Mean $ R_{T,30} $ (bps)	5-minute reaction		Drift $T+1 \rightarrow T+30$	
			β (bps/SD)	R^2	β (bps/SD)	t
ES	CPI	38.5	−20.8	0.27	−5.7	−1.4
NQ	CPI	53.6	−27.7	0.30	−4.6	−0.9
ZN	CPI	22.5	−9.9	0.30	−0.06	−0.04
YM	CPI	33.2	−19.2	0.27	−5.3	−1.4
ES	NFP	29.6	+2.8	0.03	+0.2	0.1
NQ	NFP	36.0	+2.5	0.01	−0.6	−0.4
ZN	NFP	23.6	−1.5	0.03	+0.1	0.1
YM	NFP	27.9	+2.8	0.07	+0.6	0.4
ES	PCE	14.9	−3.8	0.10	+0.4	0.2
NQ	PCE	18.5	−5.2	0.10	+1.3	0.5
ZN	PCE	8.3	−0.9	0.04	+0.5	0.5
YM	PCE	11.9	−2.9	0.09	−0.6	−0.4

Notes: β is the coefficient on the headline surprise in a regression of the return on all surprise components of the release (CPI: headline and core; NFP: payrolls, unemployment rate, average hourly earnings; PCE: headline and core), with HC3 standard errors. 122 CPI, 122 NFP and 119 PCE releases (348 with complete YM prices).

The drift regressions are weaker: with about 120 CPI releases, the coefficient on the headline surprise alone is not individually significant in any market. The drift is better measured with the family signal in equation (2), which combines all CPI lines. Its correlation with the five-minute reaction to CPI is 0.51–0.54 in all four markets, and the 30-minute drift per standard deviation of the signal is about 5.9 bps in ES, 8.5 bps in NQ, 4.4 bps in YM

and 0.8 bps in ZN. Against a five-minute reaction of 27, 37, 22 and 14 bps per standard deviation, between a fifth and a quarter of the CPI response in the equity indices remains as a drift after the first minute, and almost none in ZN. Figure 3 shows the same contrast. The trading tests in Section V, which aggregate many releases with walk-forward signs, are the decisive evidence on whether this drift is predictable out of sample.

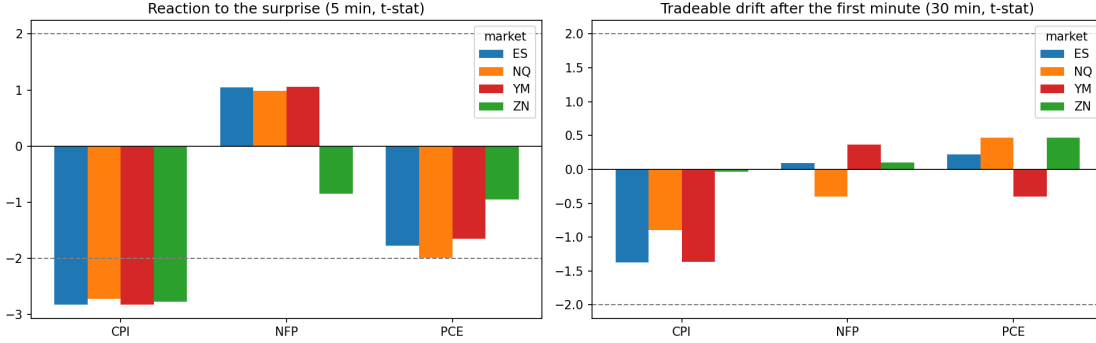


Figure 3. t -statistics of the headline surprise (CPI, payrolls, PCE) in regressions of the five-minute reaction from the pre-release price (left) and of the drift from one to 30 minutes after the release (right), ES, NQ, YM and ZN, 2016–2026. Dashed lines: ± 2 .

C Does the pre-release market state matter?

Andersen et al. (2007) show that announcement effects depend on the state of the economy. We ask a related question at high frequency: does the same surprise move prices more when markets are already stressed? We pre-register six state variables measured before the release—20-day realized volatility, the previous day’s intraday volatility, 20-day momentum, the distance from the 60-day high, the overnight return and the previous surprise of the same release—each converted to a percentile against its trailing one-year distribution, and estimate

$$r_T = a + b x_T + c s_T + d (x_T \times s_T) + \varepsilon_T, \quad (4)$$

where $s_T \in [0, 1]$ is the state percentile, so d is the change in the surprise beta from the calmest to the most stressed state. Each interaction is tested with HC3 and a 2,000-draw permutation test, corrected for the 48 primary tests with the false-discovery rate of Benjamini and Hochberg (1995), and required to have the same sign in 2016–2020 and 2021–2026.

Figure 4 summarizes the results. In ES, CPI surprises hit much harder when recent volatility is high: the interaction with 20-day volatility has t -statistics of 2.9–3.8 on the reactions at every horizon, and the five-minute beta rises by about 61 bps from the calmest to the most volatile markets. NQ shows the same pattern slightly weaker (t of 2.2–3.0), and

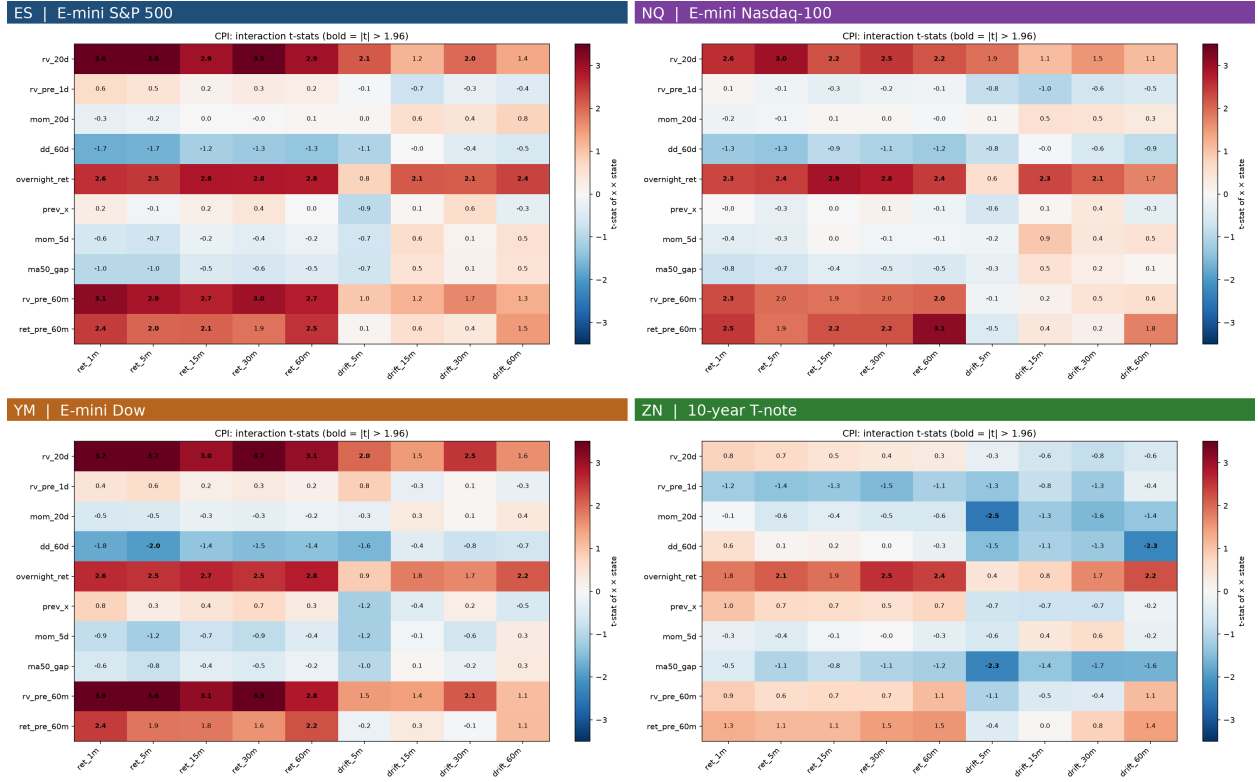


Figure 4. t -statistics of the $\text{CPI} \times \text{state}$ interaction d in equation (4), by state variable (rows) and target (columns: reactions and drifts at 1–60 minutes), for ES, NQ, YM and ZN.

YM almost exactly the same one: its five-minute CPI beta rises by 58.7 bps from calm to volatile states ($t = 3.74$, permutation $p = 0.004$), with the same sign in 2016–2020 and 2021–2026. Three features limit the economic value of this result. First, the effect is concentrated in the first-minute jump; on the tradeable drift the t -statistics fall to 1.1–2.1. Second, in walk-forward estimation the interaction was not detectable in real time before 2022: its training t -statistic was 0.5 in 2019 and below 2 until 2022. Third, only two ES tests and no NQ test survive the false-discovery correction; in YM three survive, $\text{CPI} \times \text{volatility}$ and two $\text{NFP} \times \text{volatility}$ effects that come entirely from 2016–2020. On the YM drift the CPI interaction is +24.6 bps ($t = 2.52$) but does not survive the correction. In ZN there is no volatility pattern at all. The state variables do predict the size of the move robustly: in high-volatility states the absolute 30-minute move is about 20 bps larger in ES, and 198 (ES), 154 (NQ) and 156 (YM) of 360 magnitude tests survive the correction. We return to this distinction between direction and size in Section VI.

IV Models and Evaluation

This section sets out a simple framework for the speed of price discovery, the common trading and evaluation environment, and the four forecasting models. Throughout, T denotes the release minute, P_τ the close of the one-minute bar labelled τ , $p_\tau = \ln P_\tau$, and $\mathcal{F}_T$ the information available at the close of bar T , one minute after the release. All returns are in basis points (bps).

A A partial-adjustment framework

Let v_T denote the change in the fundamental value of the asset implied by the news in a release, and suppose that it is proportional to the family signal in equation (2), $v_T = \gamma x_T$, where γ is the full-information price impact of a one-standard-deviation surprise. Prices adjust toward the new value gradually. We write the cumulative reaction after h minutes as

$$R_{T,h} = p_{T+h-1} - p_{T-1} = \Lambda_h \gamma x_T + u_{T,h}, \quad 0 \leq \Lambda_1 \leq \Lambda_2 \leq \dots \leq 1, \quad (5)$$

where Λ_h is the share of the news priced after h minutes and $u_{T,h}$ is noise unrelated to the surprise, $\mathbb{E}[u_{T,h} \mid x_T] = 0$. Under efficient and instantaneous price discovery, $\Lambda_1 = 1$. Using the decomposition in equation (3), the tradeable drift from the first post-release price is

$$D_{T,h} = R_{T,h} - R_{T,1} = (\Lambda_h - \Lambda_1) \gamma x_T + (u_{T,h} - u_{T,1}). \quad (6)$$

Equation (6) delivers the two quantities this paper measures. The first is the share of the news that is *not* priced within the first minute,

$$\theta_h \equiv 1 - \frac{\Lambda_1}{\Lambda_h} = \frac{\beta_h^D}{\beta_h^R}, \quad \beta_h^R = \frac{\text{Cov}(R_{T,h}, x_T)}{\text{Var}(x_T)}, \quad \beta_h^D = \frac{\text{Cov}(D_{T,h}, x_T)}{\text{Var}(x_T)}, \quad (7)$$

which is identified from two predictive regressions. The second is the predictability of the drift from public information: since $x_T \in \mathcal{F}_T$,

$$\mathbb{E}[D_{T,h} \mid \mathcal{F}_T] = \beta_h^D x_T, \quad (8)$$

so a strategy that takes a position in the direction of x_T at the close of bar T earns $\beta_h^D \mathbb{E}|x_T|$ per trade before costs. The drift is exploitable only if this exceeds the round-trip cost c ,

$$\beta_h^D \mathbb{E}|x_T| > c. \quad (9)$$

Condition (9) separates two reasons why a predictable drift may not be traded: the news may be priced almost entirely within a minute ($\theta_h \approx 0$), or the remaining drift may be small relative to the minimum price increment, which sets a floor on c . The average CPI paths in Section III imply $\theta_{30} \approx 0.3$ in ES, NQ and YM and $\theta_{30} \approx 0.2$ in ZN; the estimated drift slopes of about 5.9, 8.5, 4.4 and 0.8 bps per standard deviation, against round-trip costs of about 1.7, 0.7, 1.0 and 3.0 bps, satisfy condition (9) in the three equity markets and violate it in ZN.

B Trading environment

Timing and returns. Every model forms its forecast with information in $\mathcal{F}_T$ and opens a position $w_T \in [-2, 2]$ (in contracts) at the close of bar T (Figure 5). A trade on window k opens at minute $T + a_k$ and closes 30 minutes later, with gross return

$$r_{T,k}^g = w_{T,k} (p_{T+a_k+29} - p_{T+a_k-1}) \times 10^4, \quad (10)$$

with $a_1 = 1$ for the single-window models, so that $r_{T,1}^g = w_{T,1} D_{T,30}$. The last pre-release price p_{T-1} is never an entry price.

Costs. For a contract with tick size δ (in price points) and point value M (in dollars), trading n ticks of slippage per round trip plus a commission κ costs, in basis points of the

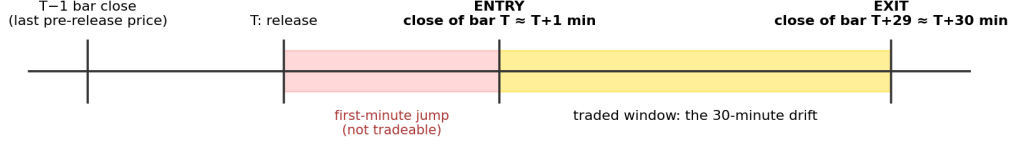


Figure 5. Timing of a trade. The release occurs at T ; the close of bar $T - 1$ is the last pre-release price; the jump is priced within bar T ; the position is opened at the close of bar T and closed at the close of bar $T + 29$.

entry price P ,

$$c(P) = \frac{n\delta + \kappa/M}{P} \times 10^4, \quad (11)$$

and the net return is $r_{T,k} = r_{T,k}^g - |w_{T,k}| c(P_{T+a_k-1})$. We report $n \in \{0, \frac{1}{2}, 1\}$ with $\kappa = 0$ (gross, a quarter and a half tick per side) and $n = 2$, $\kappa = \$4.50$ (the base case).

Performance. Let $d(t)$ be the trading session of time t (18:00–17:00 Eastern Time, dated by the close). The daily P&L of a book is $\pi_s = \sum_{(T,k): d(T+a_k)=s} r_{T,k}$, zero on sessions without a trade. Over S sessions,

$$\widehat{SR} = \sqrt{252} \frac{\bar{\pi}}{\hat{\sigma}_\pi}, \quad \text{MDD} = \min_s \left(\sum_{j \leq s} \pi_j - \max_{u \leq s} \sum_{j \leq u} \pi_j \right), \quad \hat{\beta} = \frac{\widehat{\text{Cov}}(\pi_s, \pi_s^{BH})}{\widehat{\text{Var}}(\pi_s^{BH})}, \quad (12)$$

where π_s^{BH} is the daily return of holding one contract, with roll jumps set to zero. The single-model results in Section V.A report monthly Sharpe ratios, with $\sqrt{12}$ replacing $\sqrt{252}$.

Out-of-sample estimation. Every estimated quantity—line signs, the release universe, regression slopes, gate thresholds and position-size references—is re-estimated at the start of each year y using only releases before January 1 of year y . We write $\mathcal{H}_y$ for this history. The single-model results (Section V.A) use history from 2016 and evaluate 2019–2026 at the base cost. The comparison of the four models (Sections V.C–F) uses the split in Table 2 and reports the validation years (2018–2020) and the test years (2021–2026) separately.

C Model 1: Consensus surprises

Line signs. For each market and line l , the sign $s_{l,y}$ in equation (2) is the sign of the OLS slope of the first-minute jump on the standardized surprise over $\mathcal{H}_y$,

$$R_{T,1} = a_l + b_l z_{l,T} + e_T, \quad s_{l,y} = \text{sign}(\hat{b}_l), \quad (13)$$

estimated when at least 24 releases are available.

Release universe. For each family f and year y , an impact screen compares the family's 30-minute reaction to that of other families released at the same clock time,

$$I_{f,y} = \frac{\text{mean}_{T \in \mathcal{T}_f \cap \mathcal{H}_y} |R_{T,30}|}{\text{mean}_{T \in \mathcal{C}_f \cap \mathcal{H}_y} |R_{T,30}|}, \quad (14)$$

where $\mathcal{T}_f$ are the family's release times and $\mathcal{C}_f$ the release times of other families at the family's modal clock time on which f is not released. The ten families with the largest $I_{f,y}$ (at least 30 releases each), plus CPI, form the universe $\mathcal{U}_y$.

Naive book. For every $f \in \mathcal{U}_y$ released at T ,

$$w_{f,T} = \text{sign}(x_{f,T}), \quad w_T = \frac{1}{|\mathcal{U}_T|} \sum_{f \in \mathcal{U}_T} w_{f,T}, \quad (15)$$

where $\mathcal{U}_T$ is the set of universe families released at the same minute, so no minute is traded twice.

CPI model. For CPI, the drift slope in equation (8) is estimated on the earlier CPI releases $\mathcal{H}_T = \{t < T\}$,

$$\hat{\beta}_T^D = \arg \min_b \sum_{t \in \mathcal{H}_T} (D_{t,30} - b x_t)^2, \quad \hat{r}_T = \hat{\beta}_T^D x_T, \quad (16)$$

requiring at least 24 earlier releases, and the position implements condition (9) release by release:

$$w_T = \text{sign}(\hat{r}_T) \mathbf{1}\{|\hat{r}_T| > c(P_T)\}. \quad (17)$$

D Model 2: Price-path analogues (DTW ladder)

Paths. For a trade entering at minute $e = T + a_k$, the query path is the log price over the $L + 1 = 31$ minutes ending at the entry, standardized so that only its shape matters,

$$q_j = \frac{p_{e-L+j-1} - \bar{p}}{\hat{\sigma}_p}, \quad j = 1, \dots, L + 1, \quad (18)$$

where $\bar{p}$ and $\hat{\sigma}_p$ are the mean and standard deviation of the $L + 1$ log prices.

Distance. For two standardized paths q and c of length n , the dynamic time warping distance is computed by the recursion

$$g(i, j) = (q_i - c_j)^2 + \min\{g(i-1, j), g(i, j-1), g(i-1, j-1)\}, \quad |i - j| \leq b, \quad (19)$$

with $g(0, 0) = 0$, $g(i, 0) = g(0, j) = \infty$ and $d(q, c) = g(n, n)$. The Sakoe–Chiba band $b = 5$ minutes allows the reaction to unfold up to five minutes faster or slower than in the matched past episode (Sakoe and Chiba, 1978; Berndt and Clifford, 1994).

Neighbours and forecasts. The candidate set $\mathcal{N}_e$ contains the paths of earlier releases at the same clock time and window k , within a three-year lookback, whose own trade closed before e , limited to the 600 most recent. With $\mathcal{K}_e \subset \mathcal{N}_e$ the $K = 15$ candidates with the smallest distances d_j , ages τ_j in years and realized next-30-minute returns y_j , the weights and the two forecasts are

$$\omega_j = \frac{d_j^{-1} 2^{-\tau_j/H}}{\sum_{i \in \mathcal{K}_e} d_i^{-1} 2^{-\tau_i/H}}, \quad \mu_e = \sum_{j \in \mathcal{K}_e} \omega_j y_j, \quad \sigma_e = \left(\sum_{j \in \mathcal{K}_e} \omega_j (y_j - \mu_e)^2 \right)^{1/2}, \quad (20)$$

with a recency half-life of $H = 2$ years. The direction forecast μ_e is the analogue estimate of $\mathbb{E}[D \mid \text{path}]$ and the dispersion forecast σ_e measures how much the analogues disagree.

Gate and size. With Q_α the α -quantile of μ over the fitting years and $\bar{\sigma}$ the median of σ ,

$$w_e = \text{sign}(\mu_e) \mathbf{1}\{\mu_e \leq Q_{0.3} \text{ or } \mu_e \geq Q_{0.7}\} \cdot \min \left\{ \frac{\bar{\sigma}}{\sigma_e}, 2 \right\}. \quad (21)$$

The book trades five back-to-back windows after each release of eight families, $a_k \in \{1, 31, 61, 91, 121\}$, with the forecasts in equation (20) rebuilt for each window.

E Model 3: Pattern-based event games

The event-game model applies the analogue logic to longer paths around the release. For each release T and anchor $T_0 = T + o$, $o \in \{-60, -30, 0, 30, 60\}$ minutes, the path is the log return over $[T_0, T_0 + 60]$ sampled every $\Delta = 3$ minutes and standardized as in equation (18), and the target is the next 30-minute return,

$$y_{T,o} = p_{T_0+90} - p_{T_0+60}. \quad (22)$$

Neighbours come from a pool $\mathcal{P}$ of release types (Jobless Claims, JOLTS and PCE) at the same clock time and anchor, within a two-year lookback, with a DTW band of five sampling steps. The factor is the distance- and recency-weighted average of the neighbours' targets, as in equation (20), and the book trades

$$w_{T,o} = \text{sign}(\mu_{T,o}) \quad (23)$$

one contract, entering at $T_0 + 60$. The pool, sampling frequency and lookback were chosen by a grid search over 18 combinations on the full sample.

F Model 4: Machine-learning classifier

The classifier forecasts the direction of the 30-minute return from a feature vector ϕ_T that combines surprise, price-path and momentum information. With $\hat{F}_y$ trained on $\mathcal{H}_y$,

$$\hat{\rho}_T = \Pr_{\hat{F}_y}(D_{T,30} > 0 \mid \phi_T), \quad w_T = g(\hat{\rho}_T) \in [-1, 1], \quad (24)$$

where g maps the predicted probability into a signed position that is larger when the model is more confident; we keep the model's positions exactly as produced. The *executable* version uses the trade return in equation (10) with $a_1 = 1$. The *as-reported* version enters at the last pre-release price,

$$r_T^{g,\text{rep}} = w_T (p_{T+28} - p_{T-1}) \times 10^4 = w_T (R_{T,1} + D_{T,29}) \times 10^4, \quad (25)$$

so the difference between the two isolates the value of the untradeable first-minute jump $R_{T,1}$, which by equation (5) contains the share Λ_1 of the news.

G Inference

Random-sign test. Holding the trades, sizes and costs of a book fixed, we draw $\epsilon_{T,k} \in \{-1, +1\}$ with equal probability and form $\tilde{r}_{T,k} = \epsilon_{T,k} |w_{T,k}| (p_{T+a_k+29} - p_{T+a_k-1}) \times 10^4 - |w_{T,k}| c$. With $\widehat{SR}^{(b)}$ the Sharpe ratio of the b -th of $B = 1,000$ draws,

$$p_{\text{RS}} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\widehat{SR}^{(b)} \geq \widehat{SR}\}, \quad (26)$$

which tests whether the direction adds value beyond being in the market at release times.

Block bootstrap. Confidence intervals for $\widehat{SR}$ resample blocks of three consecutive months of returns with replacement (Künsch, 1989), which preserves serial dependence in the monthly returns.

Leak test. For the analogue models we deliberately extend the query path in equation (18) five minutes past the entry. Because the extended path overlaps the target, performance must rise sharply if the timing is correct; an unchanged result would indicate that the real book already uses future prices.

H Combining the models

Let $\pi_{i,s}$ be the daily net P&L of book i at a quarter tick per side and $\hat{\sigma}_i$ its standard deviation over the validation years 2018–2020. An equal-risk portfolio of a set of books $\mathcal{B}$ has daily return

$$\pi_s^P = \sum_{i \in \mathcal{B}} \lambda_i \pi_{i,s}, \quad \lambda_i = \frac{\hat{\sigma}_i^{-1}}{\sum_{j \in \mathcal{B}} \hat{\sigma}_j^{-1}}, \quad (27)$$

and is scored on the test years only. For uncorrelated books with equal Sharpe ratios SR , the portfolio Sharpe ratio is $\sqrt{|\mathcal{B}|} SR$; the gain from combining the four models therefore depends on how correlated their daily returns are and on how much each loses after costs.

V Results

A Consensus surprises forecast the drift

We first evaluate the surprise approach on its own, walk-forward over 2019–2026 at our base cost of two ticks plus \$4.50. Trading the sign of the CPI headline surprise earns a net Sharpe ratio of 1.03 in ES and 1.00 in NQ, with 9.8 and 13.6 bps per trade over about 86 and 79

releases. The CPI model, which trades only when the predicted move exceeds the cost, earns 0.99 in ES and 0.79 in NQ, with hit rates of 64% and 58%. The same rule on the E-mini Dow, a market added after the ES and NQ results were known, earns 0.99 (77 trades, hit rate 57%). The 90% block-bootstrap confidence interval of the ES Sharpe ratio is 0.25 to 1.72. In the format common among practitioners, the ES CPI strategy gains 6.6% of notional while being long in exactly the same 30-minute windows loses 6.0%; in NQ the figures are +7.5% and -5.4%. The profit therefore comes from the direction of the surprise, not from being in the market at release times. In ZN the naive rule earns 2.7 bps per trade before costs (gross Sharpe 0.53) and -0.3 bps after them; the CPI model correctly declines to trade.

Trading the ten screened families rather than CPI alone raises the number of trades tenfold, to 98 a year in ES and 97 in NQ, with net Sharpe ratios of 0.65 and 0.71, 1.7 and 2.4 bps per trade, and random-sign p -values below 0.001 and of 0.005. CPI remains the largest contributor in both markets (+4.3% and +6.0% of notional); behind it the ranking of families differs between ES and NQ, so the broad book should be read as diversification across many small edges. On YM the broad book earns 0.68 (about 93 trades a year, random-sign $p = 0.006$). Combining ES and NQ gives about 195 trades a year and a Sharpe ratio of 0.72 (random-sign $p < 0.001$); adding YM gives about 290 trades and 0.78, with a 90% confidence interval of 0.10 to 1.50.

B Transaction costs and turnover

Figure 6 shows how the CPI strategy’s Sharpe ratio changes as the round-trip slippage rises from zero to eight ticks. The line falls gently in ES (1.15 to 0.62) and is almost flat in NQ (1.02 to 0.81), because one NQ tick is only 0.1–0.3 bps of the contract value. YM, whose tick is about 0.25–1 bp, sits between them (1.06 to 0.81). In ZN the naive rule crosses zero at about two ticks. For the broad book, the ES Sharpe ratio falls from 1.09 gross to 0.20 at four ticks, while NQ stays at 0.60 and YM falls from 0.91 to 0.45. Cost robustness is therefore a property of the market’s tick size relative to its post-release moves, not only of the strategy.

C Four approaches on equal terms

Table 4 compares the four approaches on the held-out years 2021–2026 with the same bars, entry, costs and metrics. At a quarter tick per side, the surprise book has the highest Sharpe ratio in ES (1.09), NQ (0.75) and YM (1.06), followed in ES by the classifier (0.95), the DTW ladder (0.64) and the event games (0.42). At our base cost the ranking changes: the classifier (0.81) and the surprise book (0.74) remain clearly positive, the DTW ladder falls to 0.10, and the event games, which trade about 370 times a year, turn negative. In ZN every approach

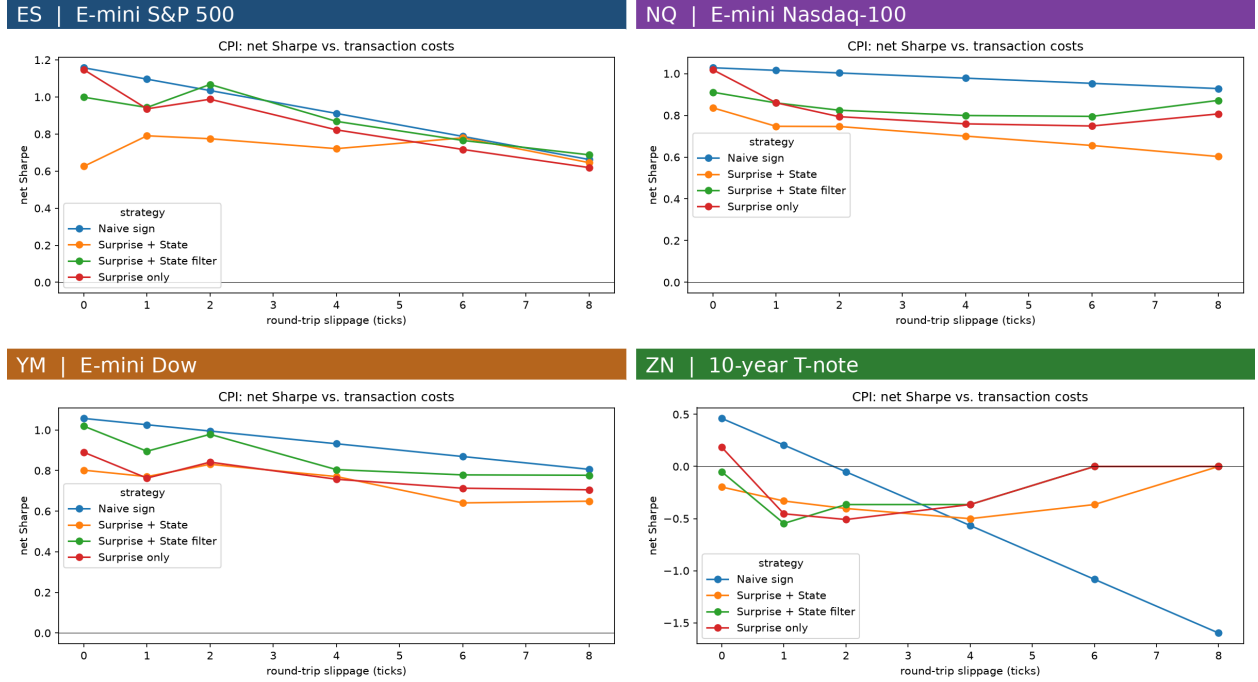


Figure 6. Net Sharpe ratio of the CPI strategies as round-trip slippage rises from 0 to 8 ticks (plus a \$4.50 commission), ES, NQ, YM and ZN, 2019–2026.

loses money after costs, although the DTW ladder earns a gross Sharpe ratio of 0.34. On YM the DTW ladder, run with the same code and setting, loses even before costs (-0.32 gross, -0.39 at a quarter tick), while the surprise book earns 1.06: the third equity index confirms the surprise and not the price-path analogues. The surprise book is positive in both the validation and the test years on every equity market (ES 1.04 and 1.09, NQ 0.86 and 0.75, YM 1.27 and 1.06).

D Combining surprises and price-path analogues

If the price path before the entry carried information beyond the surprise, conditioning the surprise on the analogues should improve it. It does not. Across all releases from 2019 to 2026, the rank correlation of the DTW direction forecast with the next 30-minute return is 0.010 in ES, 0.026 in NQ, 0.030 in YM and 0.039 in ZN, its hit rate is 47–51%, and its correlation with the surprise is 0.02–0.13. In the held-out comparison (Figure 7), keeping a surprise trade only when the analogues agree raises the ES Sharpe ratio in the validation years (1.46 against 1.04) but lowers it in the test years (0.37 against 1.09); NQ shows the same reversal (1.21 against 0.86, then 0.48 against 0.75). The reason is that the direction of the analogues itself changed sign: the DTW-only book earned 0.57 (ES) and 0.78 (NQ) in 2018–2020 and -0.26 and -0.28 in 2021–2026. YM does not show the reversal, but it

Table 4. Sharpe ratio of each strategy on each market, test years 2021–2026. “–” = the strategy is not available on that market.

Strategy	$\frac{1}{4}$ tick per side				2 ticks + \$4.50				Valid.
	ES	NQ	ZN	YM	ES	NQ	ZN	YM	ES
Macro-surprise book	1.09	0.75	−0.16	1.06	0.74	0.64	−1.51	0.82	1.04
CPI surprise model	0.51	0.46	0.17	0.39	0.61	0.45	−0.43	0.29	1.27
DTW ladder book	0.64	0.11	−0.97	−0.39	0.10	−0.04	−5.03	−0.74	0.59
DTW event-game book	0.42	–	–	–	−0.29	–	–	–	0.26
Machine-learning classifier	0.95	–	–	–	0.81	–	–	–	0.38
Buy and hold (no costs)	0.69	0.58	−0.69	0.67					0.80

Notes: Daily Sharpe ratios on the session grid. The DTW ladder uses the authors’ own feature panel and rule; the validation column reports 2018–2020 at $\frac{1}{4}$ tick per side.

does not show a gain either: its DTW-only book earns 0.40 and 0.44 in the two periods, and the agreement filter lowers the surprise book in the validation years (0.63 against 1.27) and leaves it unchanged in the test years (1.04 against 1.06). In the walk-forward evaluation on 2019–2026 a blend of the two signals raised the YM broad book from 0.68 to 1.06, but the bootstrap probability that it is not better is 6%, and the same blend does not help in ES or NQ. A filter that helps in only one of two periods, or on one of three equity indices, is not a reliable source of information. In contrast, the dispersion of the analogues predicts the size of the next move with a rank correlation of 0.10–0.30 in every market and year, and sizing surprise trades inversely to it gives Sharpe ratios of 0.63 and 0.72 in ES and 0.64 and 0.97 in NQ in the training and validation years.

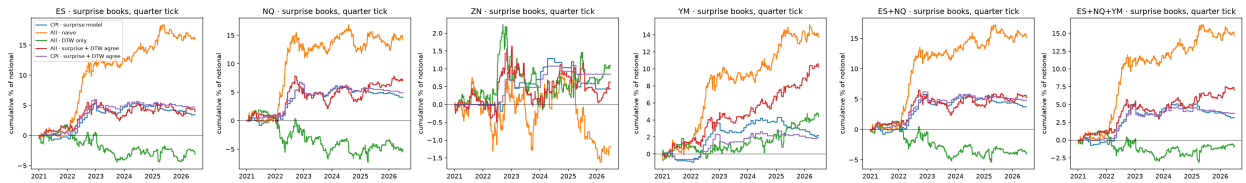


Figure 7. Cumulative net P&L at a quarter tick per side, 2021–2026, of the surprise books (top-10 families and CPI model), the DTW-only book and the surprise books filtered by DTW agreement, in ES, NQ, ZN, YM and equal-risk ES + NQ and ES + NQ + YM books.

E The entry assumption

Table 5 re-prices each approach’s own ES book on 2019–2026 at three cost levels. The classifier’s reported Sharpe ratio of 2.14 (total return 38.6% gross) falls to 0.92 (7.3%) when the same positions are entered one minute later: about 80% of the reported profit is the

first-minute jump. An independent rebuild of the classifier’s pipeline finds a loss of about 96% of the profit per trade from the same one-minute delay. The first minute is where the market prices the news, and it cannot be traded.

Table 5. Each workstream’s own book re-priced at three cost levels, ES, 2019–2026 (Sharpe / total return).

Book	Gross	$\frac{1}{4}$ tick	2 ticks + \$4.50
Macro-surprise, all releases	1.23 / 23.3%	1.12 / 21.1%	0.69 / 13.0%
DTW ladder (2018 ES rebuild) ^a	0.27 / 8.8%	0.07 / 2.4%	−0.66 / −21.5%
DTW event games	0.45 / 18.6%	0.25 / 10.3%	−0.50 / −20.6%
Classifier, executable	0.92 / 7.3%	0.88 / 6.9%	0.72 / 5.6%
Classifier, as reported	2.14 / 38.6%	2.13 / 38.3%	2.07 / 37.0%

^a Rebuilt from the published 2018 specification; the current book is in Table 4.

F Diversification across approaches

Daily returns of the four approaches are nearly uncorrelated. In the test years the correlations across approaches range from -0.04 to 0.31 , the highest being between the surprise book and the classifier, which also uses surprise features; the surprise and DTW ladder books correlate at 0.13 on ES. The high correlations are the same surprise strategy on different equity indices (ES and NQ 0.68 , ES and YM 0.85), because the same releases move all of them. We therefore combine the approaches with equal-risk weights, proportional to the inverse of each book’s daily volatility in 2018–2020, and score them on 2021–2026 (Table 6 and Figure 8). At a quarter tick per side, the ES portfolio of the four approaches earns a Sharpe ratio of **1.39**, the equity portfolio on ES, NQ and YM 1.35 and the ES + NQ portfolio 1.31 , above the best single approach (1.09) with less than half its volatility (1.2 – 1.3% against 2.7%) and a maximum drawdown of about -1% . The surprise book and the DTW ladder alone earn 1.09 and 0.64 on ES; together they earn 1.16 , because they lose in different years: 2022, the surprise book’s best year, is the only year in which the DTW ladder loses. At our base cost the equity portfolio on ES, NQ and YM holds up best (0.85), ahead of the surprise book on the same three markets (0.82) and of the ES portfolio (0.76), whose high-turnover pattern-based books pay the cost on 300 or more trades a year. Adding ZN lowers the combined Sharpe ratio to 0.83 at a quarter tick and turns it negative at our base cost.

Table 6. Equal-risk team portfolios, test years 2021–2026 (weights from 2018–2020). Return, volatility and drawdown at $\frac{1}{4}$ tick per side.

Portfolio	Sharpe				Return	Vol	Max DD	Trades/yr
	Gross	$\frac{1}{4}$ tick	$\frac{1}{2}$ tick	2t+\$4.50				
Team, ES (four strategies)	1.55	1.39	1.22	0.76	8.9%	1.2%	−1.0%	739
Team, equities (ES+NQ+YM)	1.48	1.35	1.22	0.85	9.6%	1.3%	−1.1%	1,131
Team, ES+NQ	1.45	1.31	1.17	0.78	9.0%	1.3%	−1.1%	1,043
Team, all markets (with ZN)	1.35	0.83	0.30	−1.09	4.3%	1.0%	−1.1%	1,562
Surprise + DTW ladder (ES)	1.31	1.16	1.01	0.59	13.8%	2.2%	−2.0%	315
Surprise + DTW event games (ES)	1.32	1.14	0.96	0.46	14.8%	2.4%	−2.2%	465
Macro-surprise book, ES+NQ+YM	1.13	1.07	1.01	0.82	14.6%	2.6%	−2.4%	280
Macro-surprise book, ES alone	1.18	1.09	0.99	0.74	15.9%	2.7%	−2.6%	96
DTW ladder, ES+NQ (own weights)	0.66	0.54	0.42	0.07	7.4%	2.6%	−5.3%	427

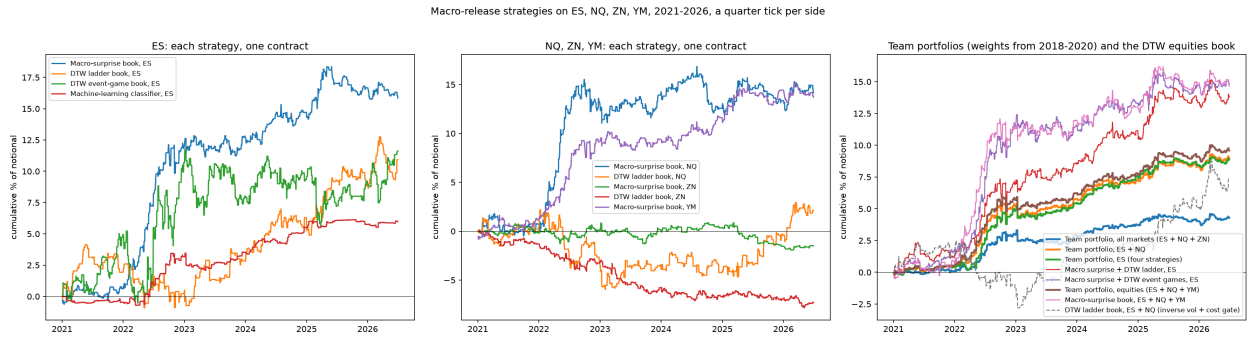


Figure 8. Cumulative net P&L at a quarter tick per side, 2021–2026: each approach in ES (left), in NQ, ZN and YM (center), and the team portfolios with the DTW ladder’s own ES + NQ book (right).

VI Interpretation

A Why equities and not Treasuries: tick size as a limit to arbitrage

The surprise explains the first-minute response of ZN as well as that of the equity indices, so the absence of a tradeable drift in ZN is not a failure of the signal. Two features of the Treasury market explain it. First, a CPI or payrolls surprise maps into the expected policy path, and hence into Treasury yields, almost mechanically, and the Treasury futures market is where the fastest participants price that information. By the time a position can be opened, the move is largely complete: the drift per standard deviation of the CPI signal is 0.8 bps in ZN, against about 6 bps in ES and 8.5 bps in NQ. For equities, investors must also judge the effect of the same number on growth, margins and risk premia, and the drift is largest in NQ, whose value depends most on long-dated cash flows; YM, with more industrial and financial stocks, reacts about 8% less than ES but drifts by the same amount after the first minute. Second, even where a gross edge exists, one ZN tick is large relative to the post-release move: the naive CPI rule earns 2.7 bps per trade before costs and pays 3.0 bps to trade, and the DTW ladder’s gross Sharpe ratio of 0.34 becomes -0.97 at a quarter tick per side. The minimum price increment thus acts as a limit to arbitrage that leaves a small predictable component in prices unexploited.

B Why price-path analogues fail on direction

Analogue forecasts assume that the mapping from recent price shapes to subsequent returns is stable. Around macro releases it was not. The analogues’ direction forecast had a negative rank correlation with subsequent returns in 2013–2017 and a positive one in 2018–2020, and the single-window analogue book that earned money in 2018–2020 lost money in ES and NQ in 2021–2026; on YM, added later, it earned little in either period. Across 567 constructions of the analogue forecast (waiting times, path lengths and neighbour weights) the training and validation correlations are almost unrelated (correlation 0.07 across settings). The change coincides with the shift of the macro regime from low inflation to the 2021–2022 inflation shock, when the same price shape could precede continuation or reversal depending on how the market read the Federal Reserve’s response. What the analogues do capture is uncertainty: when similar past paths were followed by very different outcomes, the next move tends to be large. This is why their dispersion is a stable forecast of the size of the move and a useful input for position sizing, while their direction is not.

C What the entry assumption reveals about the speed of price discovery

The classifier result in Table 5 quantifies how much of the forecastable component of returns is priced within the first minute. Moving the entry by one minute removes about 80% of the gross profit of a model that combines surprise and price information. Together with the event-study evidence that roughly a quarter of the CPI response remains after the first minute in equity futures, this places most of the price discovery within the first minute and a small, economically meaningful residual in the following half hour. The residual is predictable from public information, persists for about 30 minutes—a second 30-minute window after the release earns negative returns in every market—and is concentrated in periods when macro data drive expectations of monetary policy.

VII Robustness

A Dependence on 2022

The CPI strategy earned most of its profit in 2022, when inflation dominated markets and every CPI print moved the expected path of the policy rate (Figure 9): 79% of its net P&L in ES and 74% in NQ. Excluding 2022, its Sharpe ratio falls from 1.03 to 0.58 in ES and from 1.00 to 0.68 in NQ for the naive rule. YM repeats the pattern: 2022 contributes 4.1 of its 6.8 percentage points, and its Sharpe ratio excluding 2022 is about 0.5. The broad book, which spreads its risk over about ten releases, is less concentrated: its Sharpe ratio excluding 2022 is 0.28 in ES, 0.44 in NQ and 0.40 in YM, and the surprise books are positive in both the validation and the test years of the equal-terms comparison. We read the result as state dependence at the level of the macro regime: the drift is largest when macro data drive policy expectations.

B Entry delay and holding period

The ES CPI Sharpe ratio remains between 0.55 and 0.71 when the entry is delayed by one, two or five minutes with a 30-minute hold, and between 0.51 and 0.62 in NQ. YM is more sensitive to the delay: from 0.84 with no delay to 0.54, 0.33 and 0.17 after one, two and five minutes. Short holds are fragile: a five-minute hold loses most of its Sharpe ratio when the entry is delayed by one minute. The edge is a slow continuation over about half an hour, not a fast reaction.

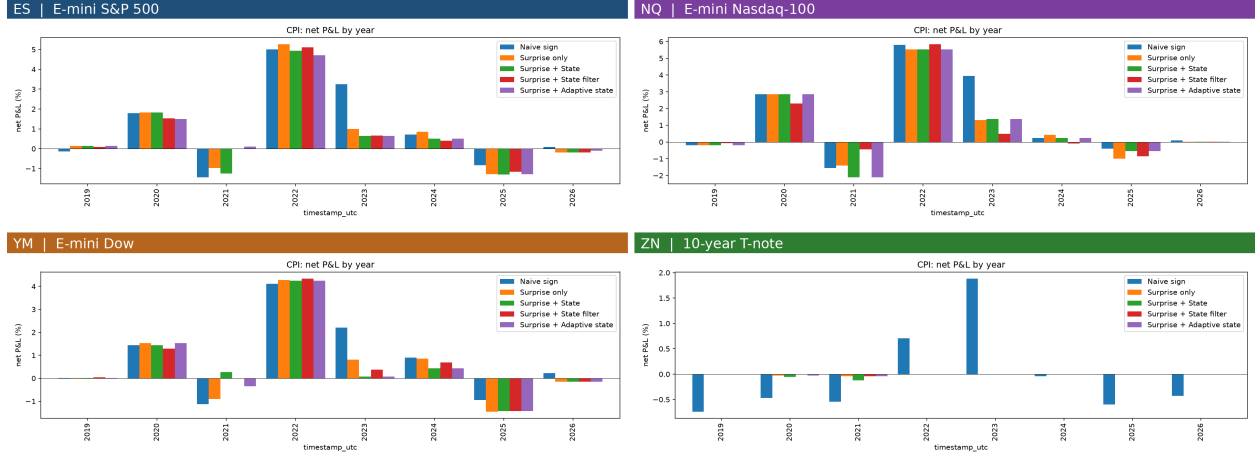


Figure 9. Net P&L of the CPI strategies by calendar year (% of notional), ES, NQ, YM and ZN.

C More windows per release

A natural way to trade more often is to trade several consecutive windows after each release. Figure 10 shows that only the first window is clearly profitable, and mainly for the surprise: the surprise’s rank correlation with the return falls from 0.139 in the first 30 minutes in ES (0.088 in NQ and 0.091 in YM) to between -0.06 and $+0.05$ in the next four windows. The analogues have no information in any window in ES, NQ and ZN; in YM their first window earns 0.47 and the later ones lose. Every additional window adds turnover and cost but not profit; five-window books lose between 29% and 48% of notional in ES over 2019–2026, and on YM the five-window surprise and DTW books have Sharpe ratios of -0.34 and -0.23 .

D Construction choices and multiple testing

Pattern-based results are sensitive to reasonable construction choices. For the ES DTW ladder, the Sharpe ratio on 2021–2026 at a quarter tick ranges from 0.17 to 0.64 across the number of neighbours, the width of the trading gate and whether its thresholds are fixed or re-estimated; across 63 variations of waiting time, path length and neighbour weights, it ranges from -0.45 to 0.61. The original implementation of the ladder, with its own family grouping and data, reaches 0.64 in ES on 2021–2026 at a quarter tick (random-sign $p = 0.04$) but 0.11 in NQ, and its direction forecast has a negative rank correlation with returns in the training years. The best of many such variations overstates what an investor could expect (Harvey et al., 2016; Bailey and López de Prado, 2014). The surprise approach has few free choices, and its main constants were fixed before the test years were scored. The pattern-based books also pass a leak test: extending the price path five minutes past the entry raises the ES DTW Sharpe ratio from 0.64 to 2.86, so the timing of the real book is correct and

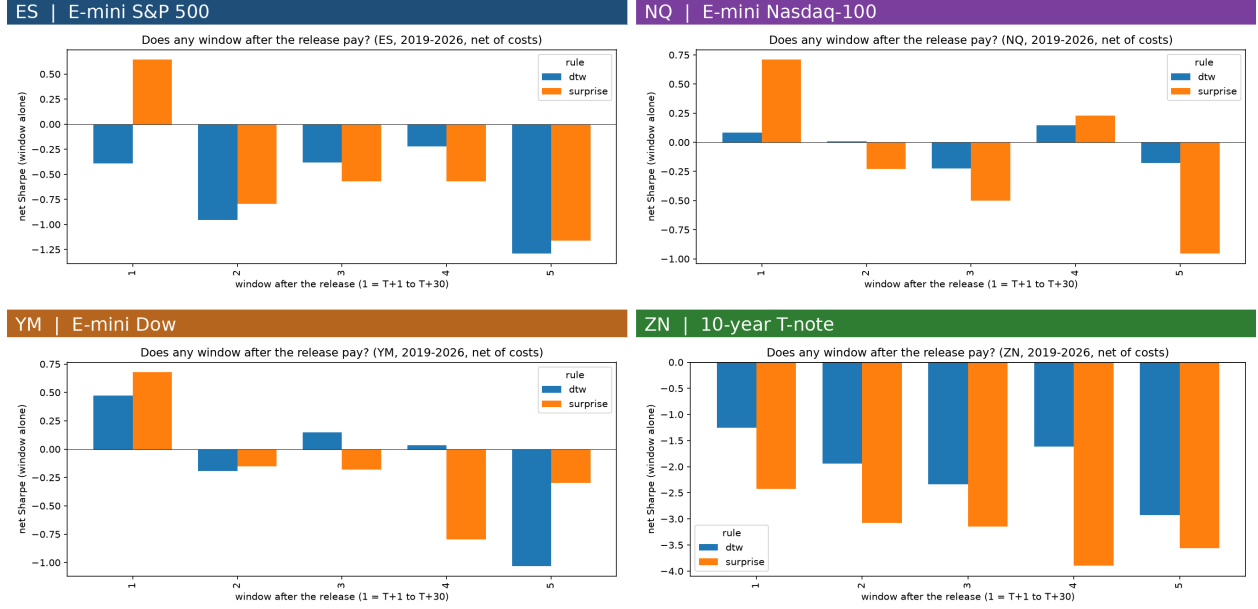


Figure 10. Net Sharpe ratio of the surprise and DTW signals traded in each of the five 30-minute windows after a release, ES, NQ, YM and ZN, 2019–2026.

the information is not coming from the future. In the state-dependence analysis, analytic HC3 and permutation p -values agree closely, and we report only effects that survive the false-discovery correction of [Benjamini and Hochberg \(1995\)](#).

E Limitations

Our results are subject to five limitations. First, costs are modeled rather than observed; fills in the first minute after a release may be worse than two ticks. Second, the sample contains about 120 releases per family, and CPI-only results rest on 50–90 trades. Third, the event-game setting was chosen on the full sample, and the YM results were produced after the ES and NQ results were known, although with unchanged code and constants. Fourth, the test years have been examined more than once during the project, so they are not an untouched hold-out for any approach. Fifth, for near-zero CPI surprises the sign depends on which CPI line is used: the headline and the family average disagree on 20 of 86 releases, which changes the CPI Sharpe ratio from 1.03 to 0.51 in ES.

VIII Conclusion

How much of the macroeconomic news is left after the first minute? In equity-index futures, about a quarter of the response to a CPI surprise arrives after the first minute, the consensus

surprise predicts its direction out of sample, and the drift lasts about 30 minutes. A strategy that trades it one minute after the release earns a Sharpe ratio of about 0.7–1.1 in ES, NQ and YM depending on the set of releases and the cost assumption, with no exposure to the market. Combined with the other approaches, which lose in different years, the equity-index portfolio reaches 1.35 at a quarter tick and 0.85 at our base cost. In Treasury futures, the surprise is priced just as reliably, but almost entirely within the first minute, and the remaining drift is smaller than one tick. Price-path analogues do not forecast the direction of the drift, although their dispersion forecasts its size, and backtests that enter at the release price attribute to skill a first-minute jump that no investor can trade.

These results have implications for both research and practice. They place most of the price discovery after scheduled releases within the first minute, identify the minimum tick as a friction that leaves a small predictable component in Treasury prices, and show that the evaluation of event-driven strategies must fix the entry at an executable price. For practitioners, the consensus surprise is the source of a small but persistent edge in equity-index futures, the dispersion of price-path analogues is a useful risk input, and execution quality in the first minute after a release decides whether the edge survives. Measuring actual fills around releases, extending the analysis to other countries and asset classes, and incorporating the text of FOMC statements are natural next steps.

References

- Andersen, T. G., Bollerslev, T., Diebold, F. X., and Vega, C. (2003). Micro effects of macro announcements: Real-time price discovery in foreign exchange. *American Economic Review*, 93(1):38–62.
- Andersen, T. G., Bollerslev, T., Diebold, F. X., and Vega, C. (2007). Real-time price discovery in global stock, bond and foreign exchange markets. *Journal of International Economics*, 73(2):251–277.
- Bailey, D. H. and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107.
- Balduzzi, P., Elton, E. J., and Green, T. C. (2001). Economic news and bond prices: Evidence from the U.S. treasury market. *Journal of Financial and Quantitative Analysis*, 36(4):523–543.
- Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300.
- Berndt, D. J. and Clifford, J. (1994). Using dynamic time warping to find patterns in time series. In *AAAI Workshop on Knowledge Discovery in Databases*, pages 359–370.
- Fleming, M. J. and Remolona, E. M. (1999). Price formation and liquidity in the U.S. treasury market: The response to public information. *Journal of Finance*, 54(5):1901–1915.
- Gürkaynak, R. S., Sack, B., and Swanson, E. T. (2005). Do actions speak louder than words? The response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3):1217–1241.
- Lucca, D. O. and Moench, E. (2015). The pre-FOMC announcement drift. *Journal of Finance*, 70(1):329–371.

- MacKinnon, J. G. and White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3):305–325.
- Sakoe, H. and Chiba, S. (1978). Dynamic programming algorithm optimization for spoken word recognition. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 26(1):43–49.
- Savor, P. and Wilson, M. (2013). How much do investors care about macroeconomic risk? Evidence from scheduled economic announcements. *Journal of Financial and Quantitative Analysis*, 48(2):343–375.

Appendix A Surprise Construction

Release families. Two release lines belong to the same family if the share of the larger line’s release timestamps that are shared by both is at least 80%; families are formed transitively (union-find). The family is named after its best-known line (e.g. “CPI”). Families need at least 30 releases to be traded.

Standardization. For each line, $z_{l,t}$ in equation (1) uses the standard deviation of the line’s earlier raw surprises only, requires at least 24 of them and is capped at ± 5 . Releases are de-duplicated by line and timestamp, keeping the last revision.

Signs and screen. For each market and year, the sign s_l is the sign of the OLS slope of the first-minute return on $z_{l,t}$ over earlier years (at least 24 observations). The impact screen ranks families by the ratio of their mean absolute 30-minute reaction to the same statistic at the same clock time on other families’ releases, over earlier years, and admits the top ten, plus CPI.

Appendix B Dynamic Time Warping Forecast

For a query path q and a candidate path c , both z-scored and of length n , the DTW distance is

$$d(q, c) = \min_{\pi \in \mathcal{P}_w} \sum_{(i,j) \in \pi} (q_i - c_j)^2,$$

where $\mathcal{P}_w$ is the set of monotone alignment paths that stay within a Sakoe–Chiba band of w minutes of the diagonal (Sakoe and Chiba, 1978). Candidates are earlier releases at the same clock time whose trade had closed before the current entry, within a three-year lookback and at most the 600 most recent. For the k nearest candidates with distances d_j , ages a_j (years) and subsequent 30-minute returns y_j , the weights are $\omega_j \propto d_j^{-1} 2^{-a_j/2}$, the direction forecast is $\mu = \sum_j \omega_j y_j$ and the dispersion forecast is $\sigma = (\sum_j \omega_j (y_j - \mu)^2)^{1/2}$. The ladder trades $\text{sign}(\mu)$ when μ lies outside the 30th–70th percentiles of its earlier values, with size $\min\{\text{median}(\sigma)/\sigma, 2\}$ contracts.

Appendix C Additional Figures and Tables

This appendix collects supporting results from the project’s notebooks 02–07, in the order of the analysis: the event study (notebook 02), state dependence (03), the walk-forward CPI strategy (04), the broad surprise book (05), surprise and DTW (06) and the multi-market book (07). Unless noted, the evaluation window is 2019–2026, the cost is two ticks plus \$4.50 per round trip, and every estimate uses only earlier data.

A Event study (notebook 02)

Figure A1 shows, for ES, how large the move after CPI, payrolls and PCE is at each horizon and how much of it the surprise explains. The move after CPI keeps growing for about 30 minutes; the share explained by the surprise is highest at one minute and stays near 0.25, while payrolls move ES almost as much but with an R^2 below 0.07. Figure A2 shows the coefficient of each headline surprise by horizon: a one-standard-deviation hot CPI print lowers ES by 18 bps after one minute and by about 24 bps after 30 minutes. Table A1 reports the mean absolute move in all four markets.

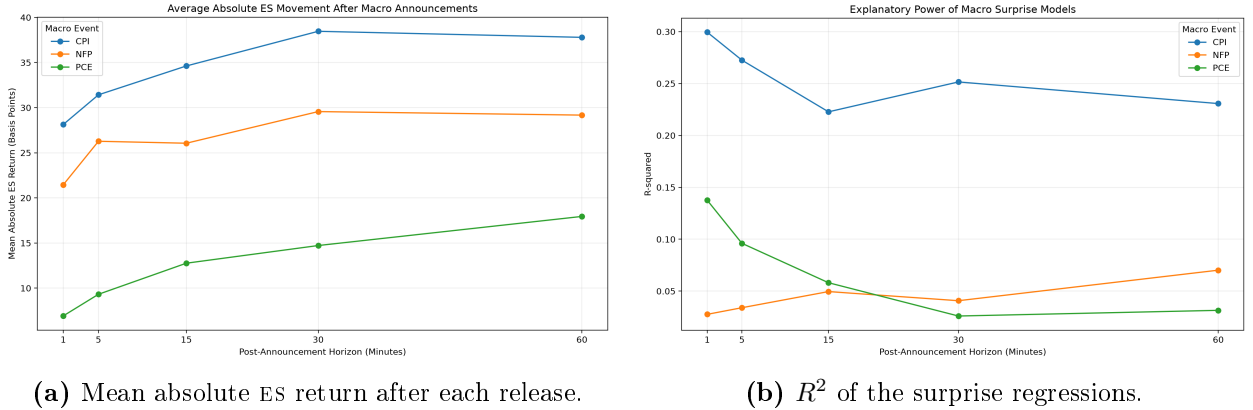


Figure A1. ES after CPI, payrolls (NFP) and PCE, 2016–2026, by horizon.

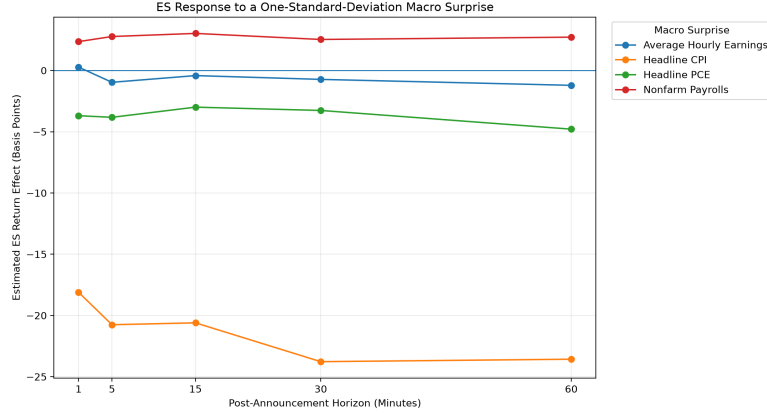


Figure A2. ES response to a one-standard-deviation surprise in each headline statistic, by horizon (bps), from regressions on all surprise components of the release.

Table A1. Mean absolute return after the release (bps), 2016–2026.

Market	Release	1 min	5 min	15 min	30 min	60 min
ES	CPI	28.1	31.4	34.6	38.5	37.8
	NFP	21.5	26.3	26.1	29.6	29.2
	PCE	7.0	9.4	12.9	14.9	18.1
NQ	CPI	38.6	44.7	48.4	53.6	54.0
	NFP	25.6	31.6	31.5	36.0	35.1
	PCE	9.2	12.5	17.2	18.5	23.2
YM	CPI	23.7	27.7	29.6	33.2	33.3
	NFP	19.3	23.7	25.0	27.9	28.6
	PCE	5.4	7.3	10.2	11.9	15.9
ZN	CPI	18.4	20.2	21.4	22.5	24.0
	NFP	20.0	21.6	22.2	23.6	24.4
	PCE	4.9	6.1	7.3	8.3	9.3

B State dependence (notebook 03)

Table A2 lists the six pre-registered state variables, all measured before the release, with the hypothesis each was chosen to test. Four further variables are reported in the figures but were not used to select models: 5-day momentum, the gap to the 50-day moving average, realized volatility over the last 60 minutes, and the return over the last 60 minutes.

Figure A3 shows three of the daily states in each market from 2010 to 2026, computed on roll-adjusted daily closes. The states vary a lot over the sample, so the terciles used below compare genuinely different regimes: 20-day volatility in the equity indices is near 10% in the quiet years before 2018 and peaks at 90–100% in March 2020, with further spikes in 2011, 2015, 2018, 2022 and April 2025, and drawdowns from the 60-day high reach –35 to –40% in 2020. ZN’s states move with the rate cycle rather than with equity stress: its volatility is

Table A2. Pre-registered state variables and hypotheses.

State	Definition (known before T)	Hypothesis
<code>rv_20d</code>	20-day realized volatility of daily closes (annualized %)	High-volatility regimes give a larger surprise beta (uncertainty, thinner liquidity)
<code>rv_pre_1d</code>	Realized volatility of 5-minute returns over the last ~ 23 trading hours (bps)	Short-term stress amplifies the reaction
<code>mom_20d</code>	20-day log return (%)	After strong rallies, bad news hits harder (stretched positioning)
<code>dd_60d</code>	Distance from the 60-day high (%)	In drawdowns the market is more sensitive to inflation news
<code>overnight_ret</code>	Return from the prior 16:00 ET close to $T - 1$ (bps)	A large pre-release move signals partial pre-positioning, so less is left to react
<code>prev_x</code>	Previous signed surprise of the same family	Surprise streaks: a second hot print confirms the trend, so the response is larger

Exploratory (reported, not used for selection): `mom_5d`, `ma50_gap`, `rv_pre_60m`, `ret_pre_60m`. Each state enters equation (4) as a percentile against its trailing one-year distribution.

elevated through 2022–2023 and its deepest drawdown (-10%) is in 2022. The ES volatility of about 85% in 2011 is overstated: the notebook computes it from the original vendor file, whose early sessions are incomplete; YM, from a complete file, peaks at about 45% in the same episode.

Figure A4 splits the five-minute CPI beta by tercile of each pre-release state in ES and YM. The beta rises from under 10 bps per standard deviation in the calmest third of 20-day volatility to 40–70 bps in the middle and top thirds in both markets, the clearest pattern among the six states. Table A3 lists the strongest of the 48 primary tests in each market. Figure A5 shows that the analytic HC3 and permutation p -values agree closely, so the inference does not depend on the normal approximation.

Table A3. Strongest state interactions among the 48 primary tests, 2016–2026.

Market	Interaction	Target	d (bps)	t (HC3)	p (perm.)	q (BH)
ES	CPI $\times$ 20-day vol	5-min reaction	+61.1	3.81	0.000	0.010
	NFP $\times$ 20-day vol	30-min drift	−12.2	−2.99	0.016	0.082
	All $\times$ overnight return	30-min drift	+7.4	2.65	0.021	0.134
YM	CPI $\times$ 20-day vol	5-min reaction	+58.7	3.74	0.004	0.014
	NFP $\times$ previous-day vol	5-min reaction	−21.0	−3.07	0.012	0.042
	NFP $\times$ 20-day vol	5-min reaction	−17.3	−3.11	0.014	0.042

Notes: d is the change in the surprise beta from the calmest to the most stressed state in equation (4). q : Benjamini–Hochberg over the 48 primary tests per market. The NFP effects in YM come entirely from 2016–2020.

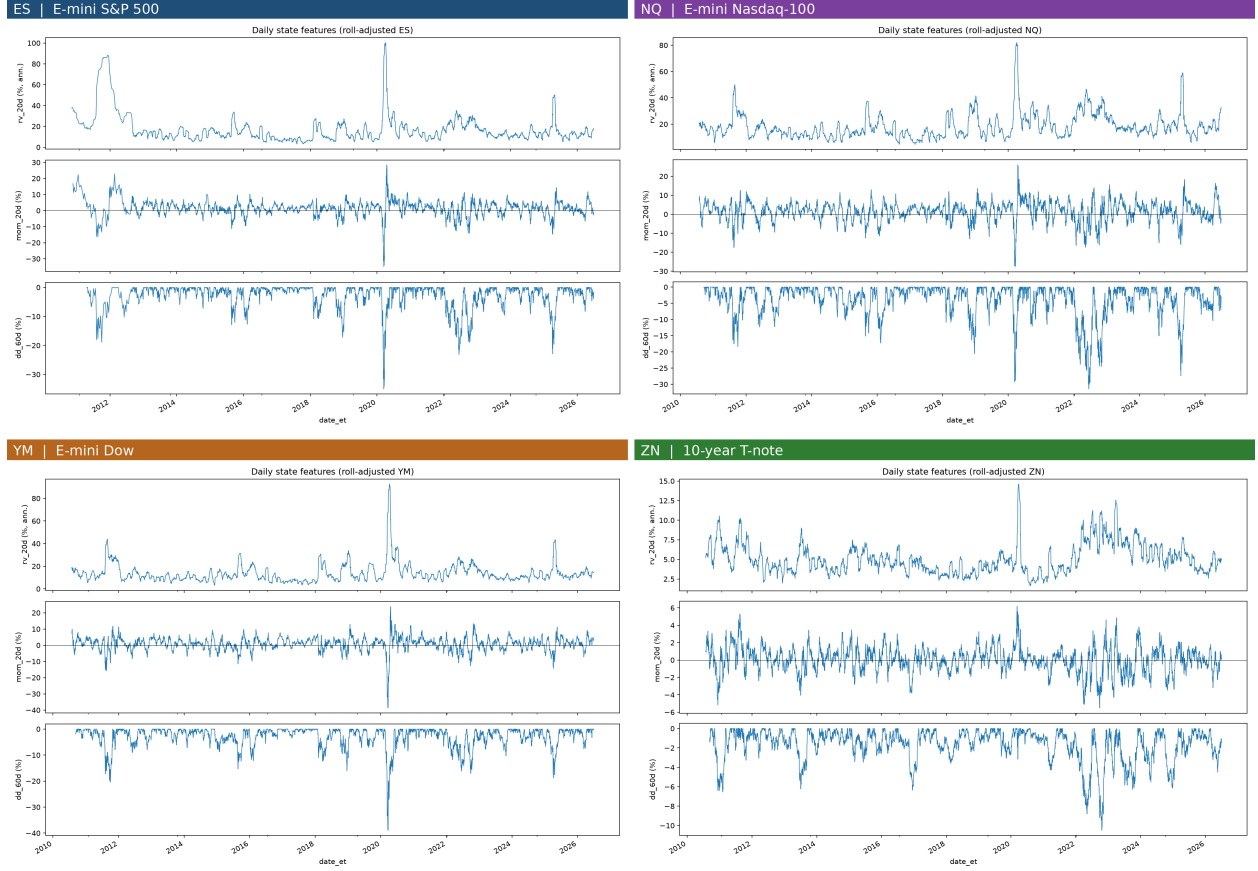


Figure A3. Daily state features, 2010–2026: 20-day realized volatility (annualized %), 20-day momentum (%) and distance from the 60-day high (%), computed on roll-adjusted daily closes. The 2011 ES volatility spike reflects incomplete sessions in the original vendor file.

Interaction regressions (notebook 03, section 9). Rather than splitting the releases into terciles, the interaction regression uses the full sample at once. For release i ,

$$r_i = a + b x_i + c s_i + d (x_i \times s_i) + \varepsilon_i, \quad s_i = \text{pct}_i - 0.5, \quad (28)$$

where x_i is the signed surprise and pct_i the percentile of the state against its trailing one-year distribution. Centering the state makes b the surprise beta at the median state; c asks whether the state predicts the return on its own; and d , the key coefficient, is the change in the surprise beta from the lowest to the highest state percentile. Because x is signed so that $b > 0$, a positive d means a stronger reaction in the high state. The pooled model (ALL) gives each family its own intercept and slope but a common c and d . Each market has 360 regressions: ten states (six primary, four exploratory), four families (CPI, NFP, PCE and pooled) and nine targets (the reaction from the pre-release price at 1–60 minutes and the drift from one minute after the release at 5–60 minutes). Standard errors are HC3.

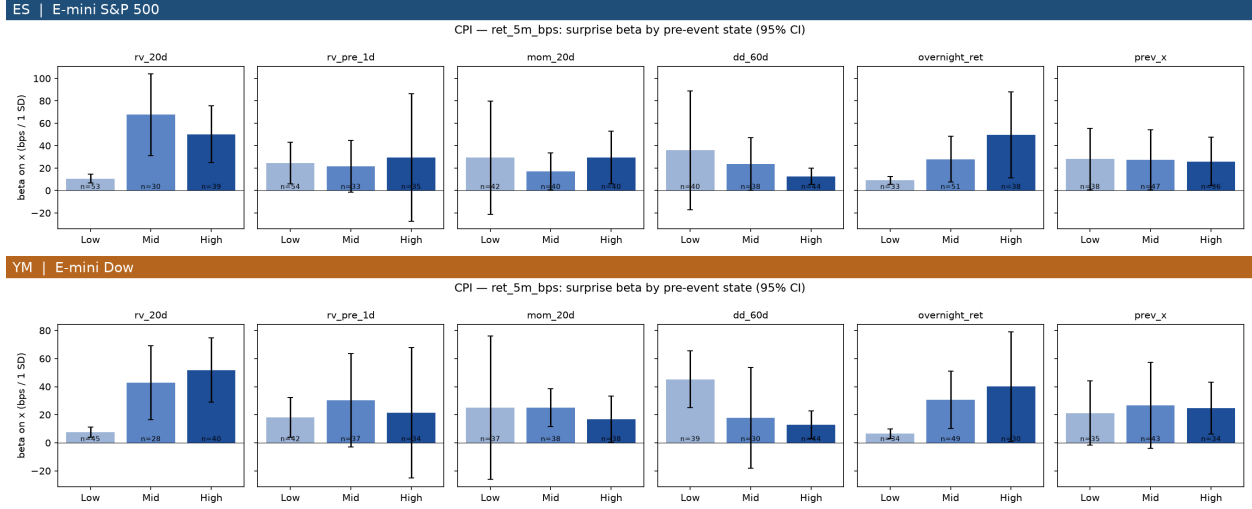


Figure A4. Five-minute CPI surprise beta (bps per SD, 95% intervals) by tercile of each pre-release state: 20-day realized volatility, previous-day volatility, 20-day momentum, 60-day draw-down, overnight return and previous surprise.

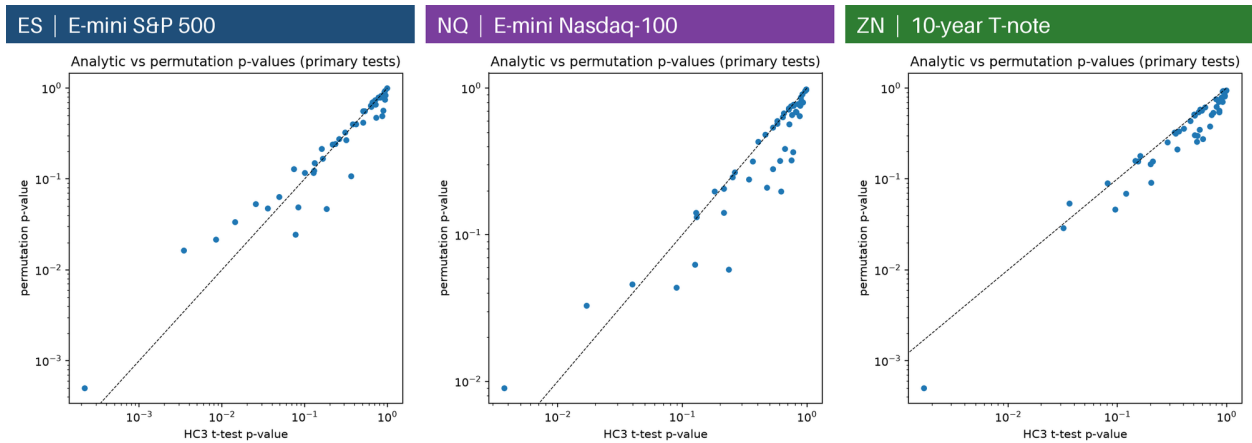


Figure A5. Analytic (HC3) against permutation p -values for the primary state tests, ES, NQ and ZN.

The 48 primary tests (six states, four families, two targets: the five-minute reaction and the 30-minute drift) also get a 2,000-draw permutation test and the Benjamini–Hochberg correction. A credible state effect looks like a row of same-colored cells across horizons; a single bold cell among about 90 is what chance produces, since at 5% significance four to five false positives are expected per heat map.

The CPI heat maps are in Figure 4 of the main text. Figures A6–A8 show the other families. For payrolls (Figure A6), the volatility rows are negative: strong payrolls help equities less when volatility is high. The pattern is strongest in YM, where both 20-day and previous-day volatility have t -statistics of -3.1 to -4.5 on the 5- to 60-minute reactions, and

present in ES on the drifts; but in YM it comes entirely from 2016–2020, so it is not used for trading. PCE (Figure A7) shows only scattered cells with no consistent row, as expected for a release whose surprise explains little of the move. In the pooled model (Figure A8), the overnight return and 5-day momentum rows are positive on the 15–60-minute drifts in ES and YM (t of 2.1–2.9): the drift follows the surprise more strongly after a market has already been rising. Only the ES overnight-return effect on the 30-minute drift is a primary test, and it does not survive the correction ($q = 0.13$). In ZN, the previous-day volatility row is negative on every drift (t of -2.1 to -3.2): after volatile days, the little drift ZN has is even smaller.

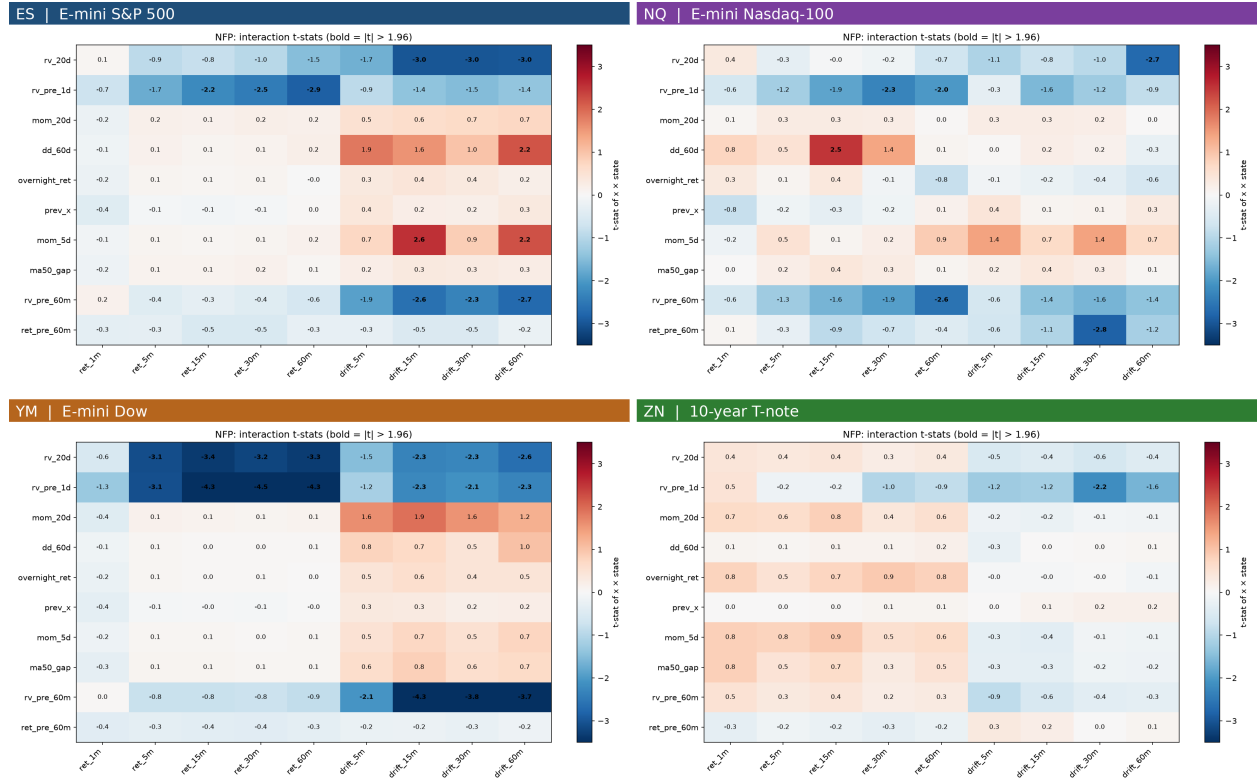


Figure A6. Payrolls (NFP): t -statistics of the surprise $\times$ state interaction d in equation (28), by state (rows) and target (columns). Bold: $|t| > 1.96$.

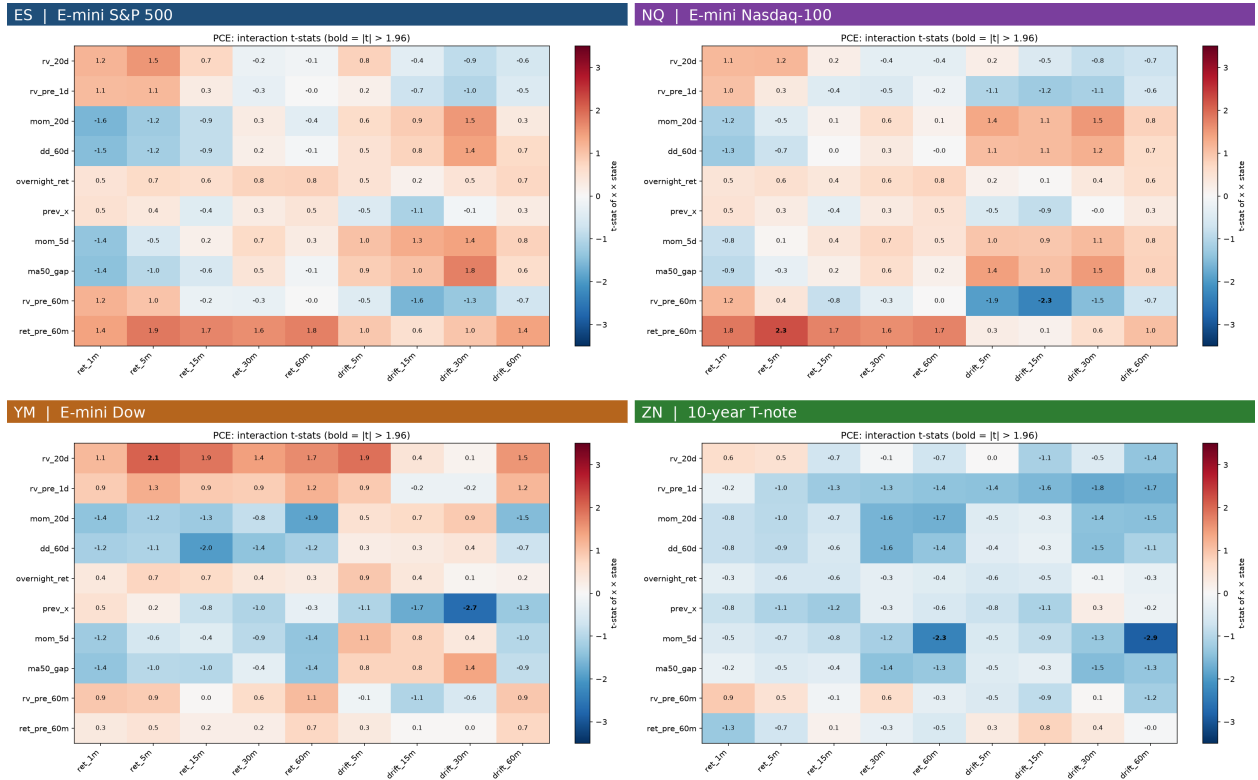


Figure A7. PCE: t -statistics of the surprise $\times$ state interaction d , by state and target.

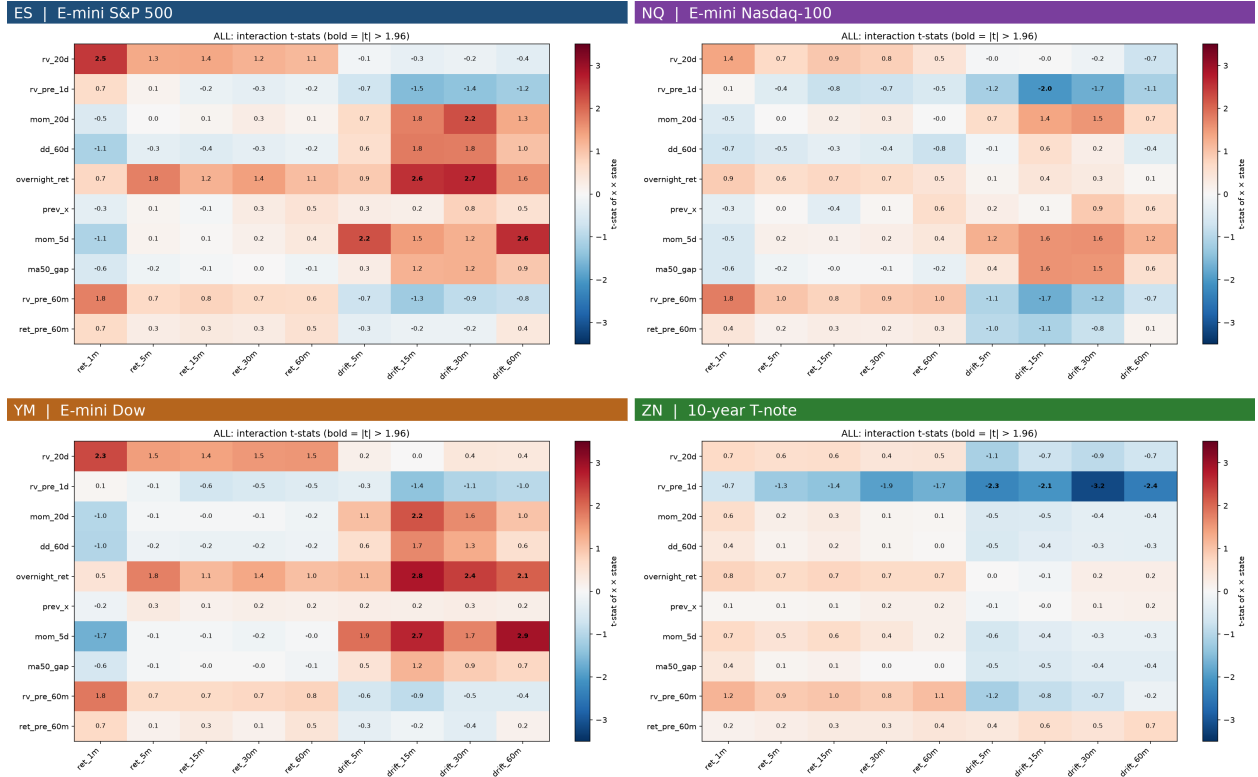


Figure A8. All three families pooled (family-specific intercepts and slopes, common c and d): t -statistics of the surprise $\times$ state interaction, by state and target.

C Walk-forward CPI strategy (notebook 04)

Figure A9 shows how the CPI strategy’s Sharpe ratio changes with the entry delay and the exit time. In both ES and YM the 30-minute exit is the best column at every delay, and short holds lose most of their value when the entry is delayed. Figure A10 shows the walk-forward estimates behind the state model: the CPI drift slope in ES fell to about 1 bp per standard deviation in 2021 and rose to 6–8 bps from 2023, and the t -statistic of the volatility interaction stays below about 2 until 2022 in ES and NQ and never reaches it in ZN. Figures A11–A13 show the walk-forward strategies in the peer chart format (notebook 04, section 16.5): the strategy before and after costs against being always long, one contract, in exactly the same 30-minute windows. In the three equity indices the strategies and the always-long line diverge from 2022: trading the CPI surprise gains 6.6% in ES, 7.5% in NQ and 5.0% in YM while always long in the same windows loses 5–6%, so the profit comes from the direction of the surprise, not from being in the market at release times. The naive sign rule and the model on CPI, PCE and payrolls show the same pattern. In ZN the surprise model trades only a handful of releases, because its predicted drift rarely exceeds the cost of one ZN tick, and the naive rule loses after costs.

Final comparison. Table A4 and Figure A14 compare the five CPI strategies of notebook 04 (section 14) in all four markets and three release universes. In the equity indices every CPI strategy has a positive Sharpe ratio, and the naive sign rule is the best or close to it (1.03 in ES, 1.00 in NQ, 0.99 in YM); conditioning on the pre-release state never improves on the surprise alone by a significant margin (the smallest bootstrap probability that a state model is not better is 0.23). Adding PCE and payrolls lowers the naive rule in ES and NQ but less so in YM. In ZN the fitted models almost never trade, because their predicted drift rarely exceeds one tick, and every strategy is negative.

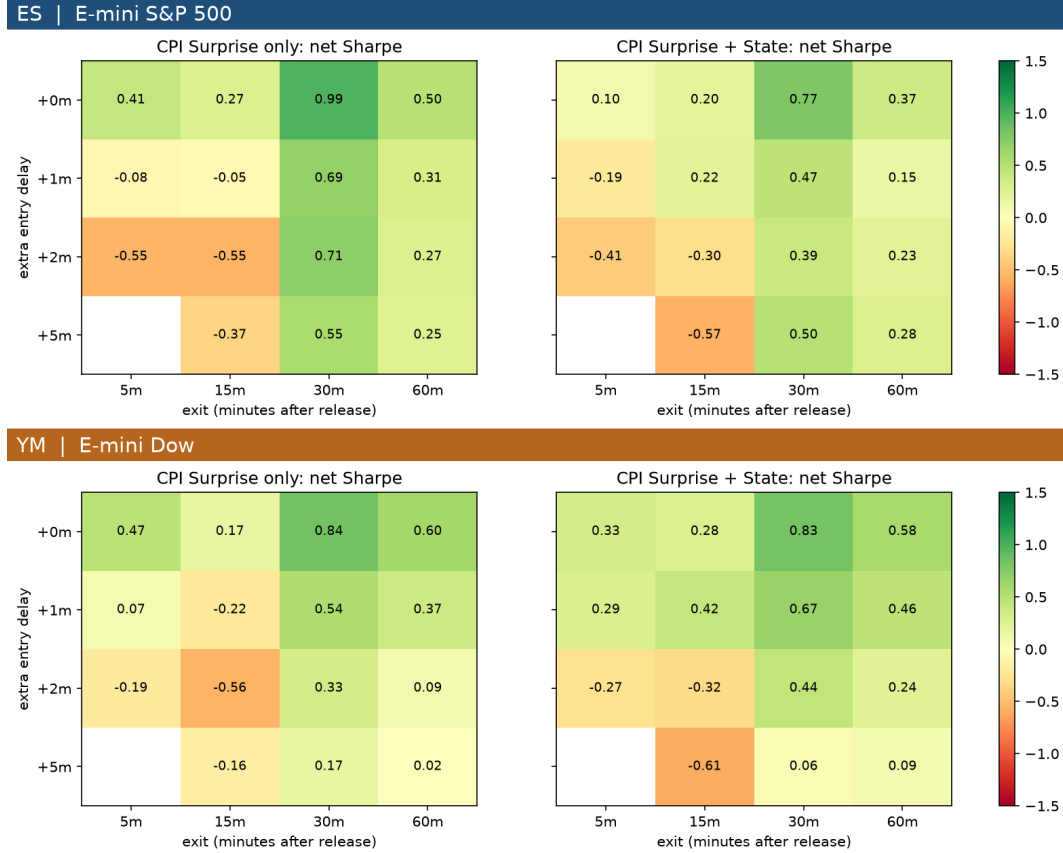


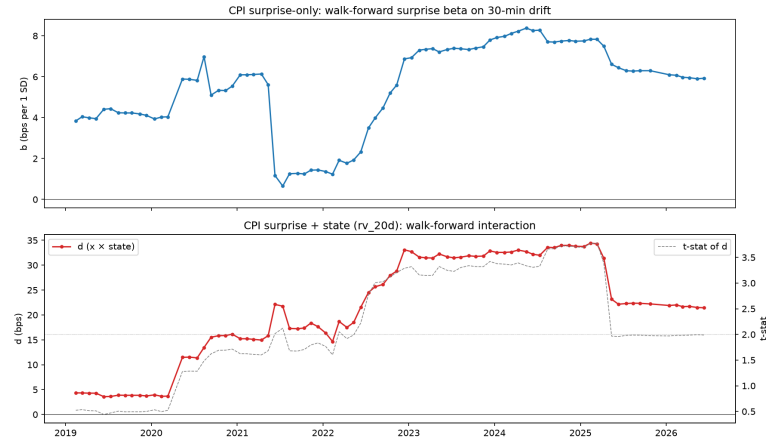
Figure A9. Net Sharpe ratio of the CPI strategies by extra entry delay (rows) and exit time (columns), 2019–2026. Left: surprise only; right: surprise and state.

Table A4. Walk-forward CPI strategies, net Sharpe ratio (90% block-bootstrap interval), 2019–2026, 2 ticks + \$4.50.

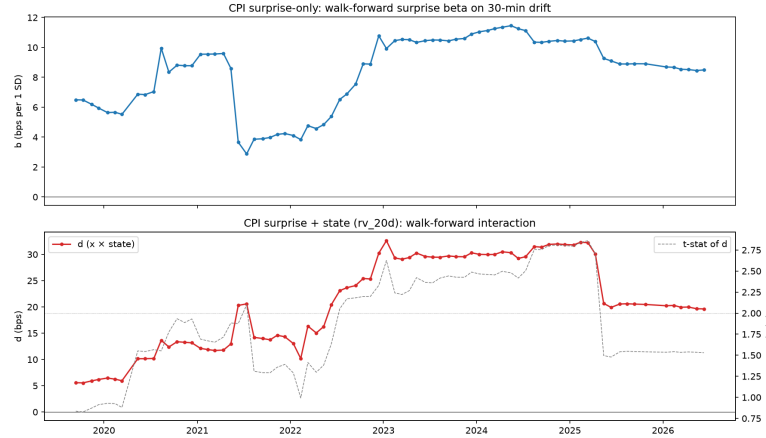
Strategy	ES	NQ	YM	ZN
<i>CPI only</i>				
Naive sign	1.03 [0.32, 1.76]	1.00 [0.37, 1.66]	0.99 [0.27, 1.72]	−0.05 [−0.82, 0.56]
Surprise only	0.99 [0.25, 1.72]	0.79 [0.12, 1.47]	0.84 [0.07, 1.61]	−0.51 (2 trades)
Surprise + State	0.77 [0.01, 1.51]	0.75 [0.04, 1.44]	0.83 [0.13, 1.51]	−0.40 (5 trades)
Surprise + State filter	1.07 [0.40, 1.67]	0.82 [0.20, 1.38]	0.98 [0.25, 1.65]	−0.37 (1 trade)
Surprise + Adaptive state	0.95 [0.26, 1.61]	0.75 [0.04, 1.44]	0.74 [0.02, 1.44]	−0.51 (2 trades)
<i>CPI, PCE and payrolls</i>				
Naive sign	0.33	0.38	0.57	−0.65
Surprise only	0.59	0.37	0.78	−0.22
Surprise + State filter	0.65	0.37	0.81	−0.07
CPI trades, naive / surprise only	86 / 52	79 / 62	77 / 48	86 / 2

Notes: Entry at $T + 1$, exit at $T + 30$; every slope, sign and state model re-estimated on earlier releases only. Intervals from a three-month block bootstrap.

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

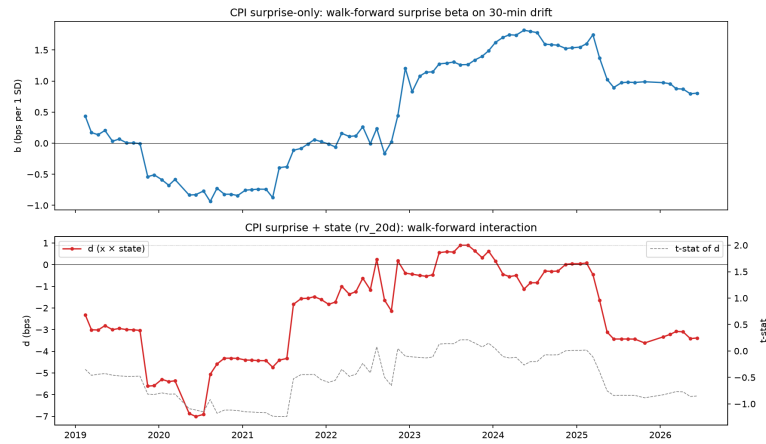


Figure A10. Walk-forward estimates for CPI, re-estimated before each release: the surprise slope and the surprise $\times$ 20-day volatility interaction with their t -statistics, ES, NQ and ZN.

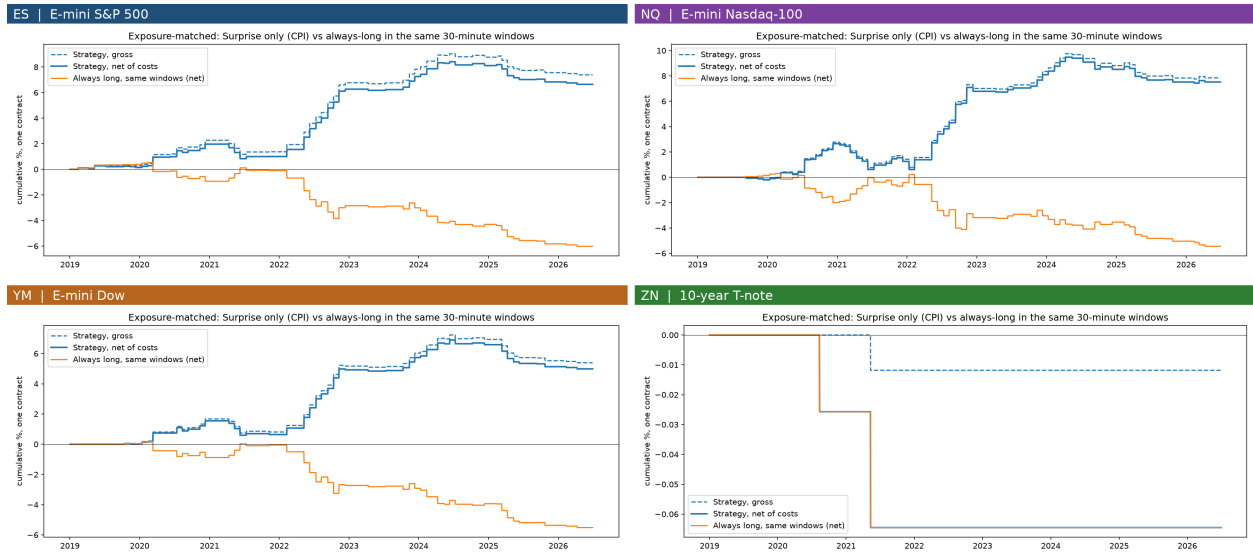


Figure A11. Peer chart format, CPI surprise model, 2019–2026: cumulative return of one contract, gross and net of costs, against always long in the same windows.

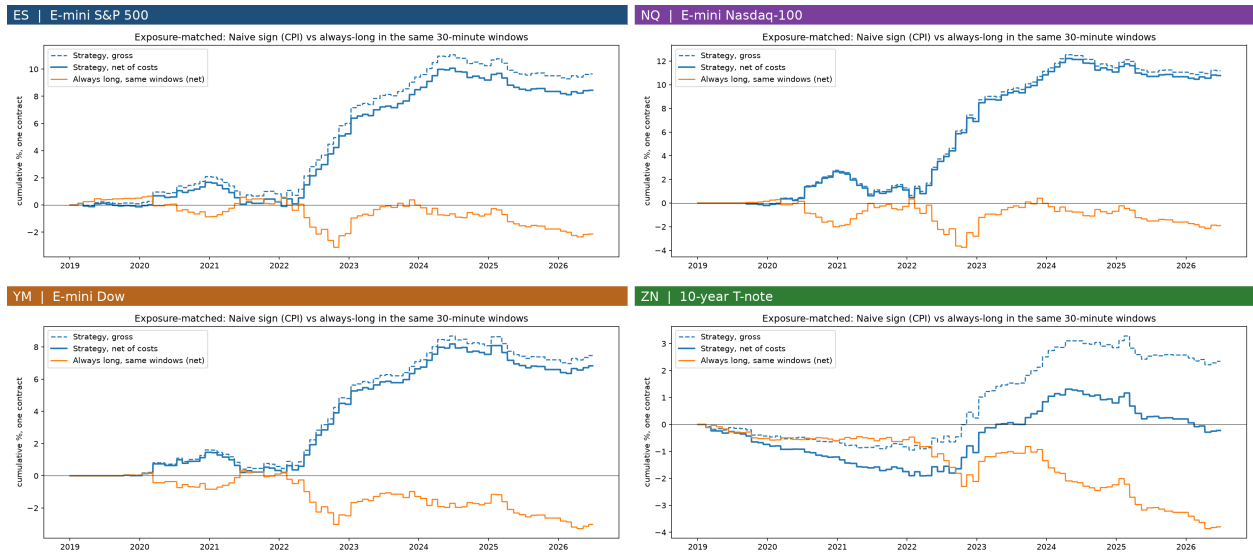


Figure A12. Peer chart format, CPI naive sign rule (every CPI release, sign of the headline surprise), 2019–2026.

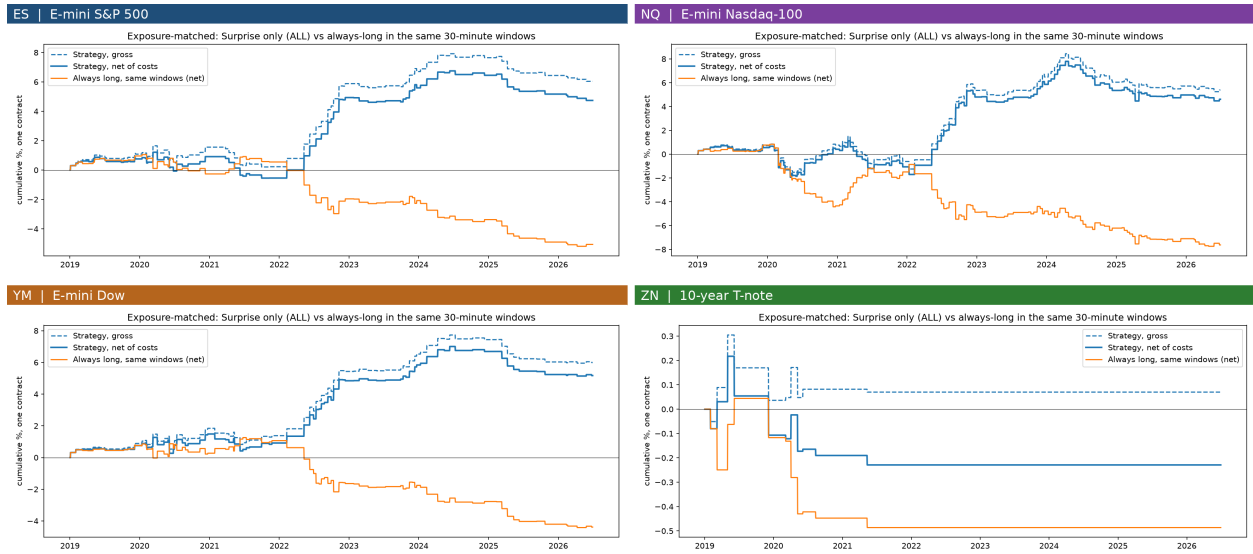


Figure A13. Peer chart format, surprise model on CPI, PCE and payrolls, 2019–2026.

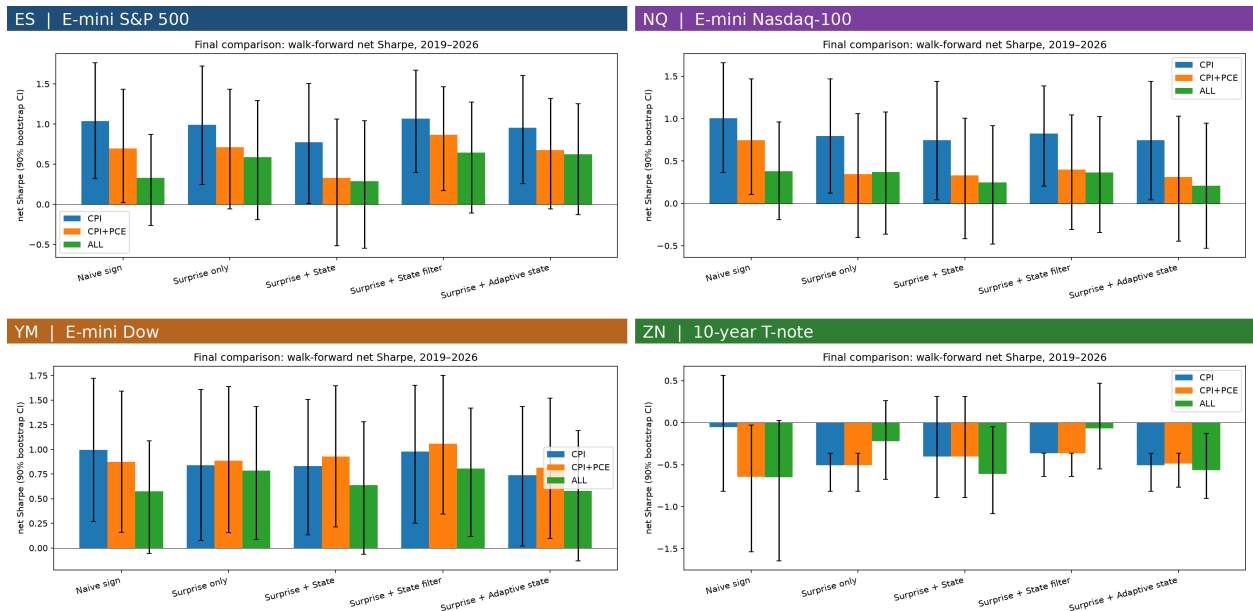


Figure A14. Final comparison of notebook 04: walk-forward net Sharpe ratio with 90% bootstrap intervals by strategy and release universe (CPI; CPI + PCE; CPI + PCE + NFP), 2019–2026.

D Broad surprise book (notebook 05)

Which releases earn money? (notebook 05, section 9.2). Table A5 and Figure A15 split the broad book’s net P&L (naive sizing, 2019–2026) by release family. CPI is the largest contributor in every equity index: +4.3% of notional in ES, +6.0% in NQ and +3.8% in YM. Behind it the ranking differs by market. In ES the next contributors are JOLTS (+3.0%), retail sales (+2.3%) and the University of Michigan survey (+1.4%); in NQ factory orders (+4.6%), the University of Michigan survey (+4.2%), retail sales (+2.6%) and GDP (+2.1%); in YM ISM Manufacturing (+3.3%), JOLTS (+2.3%) and retail sales (+1.7%). Payrolls, the release with the largest market impact, earns nothing or loses in all three (−1.7%, −0.5% and −0.1%), and ISM Services, the regional Fed surveys and pending home sales lose small amounts. No major family is individually significant: the t -statistics of the main families are at most 1.9 with naive sizing, and CPI with volatility sizing reaches 2.0 in ES and 1.9 in NQ. The broad book therefore works through diversification across many small edges rather than through one release. In ZN every family with more than a handful of trades loses, led by payrolls (−4.9%), retail sales (−4.5%) and CPI (−3.0%): its hit rates of 22–47% reflect a drift smaller than the cost.

Table A5. Net P&L of the broad surprise book by release family, 2019–2026: total net return (% of notional) and t -statistic of the mean trade, naive sizing.

ES			NQ			YM		
Family	Total	t	Family	Total	t	Family	Total	t
CPI	4.3%	1.40	CPI	6.0%	1.50	CPI	3.8%	1.48
JOLTS	3.0%	1.56	Factory Orders	4.6%	1.89	ISM Manuf.	3.3%	1.02
Retail Sales	2.3%	1.12	U. of Michigan	4.2%	1.45	JOLTS	2.3%	1.19
U. of Michigan	1.4%	0.67	Retail Sales	2.6%	1.13	Retail Sales	1.7%	1.01
Adv. Goods Trade	1.3%	1.13	GDP	2.1%	1.55	Trade Balance	1.2%	0.88
ISM Services	−1.5%	−0.49	ISM Services	−1.3%	−0.32	Wholesale Trade	−1.6%	−2.40
Payrolls (NFP)	−1.7%	−0.60	Richmond Fed	−1.9%	−1.02	Wholesale Inv.	−2.7%	−1.68

Notes: Top five contributors and the two largest losers in each market. Payrolls: −0.5% in NQ and −0.1% in YM. In ZN every family with more than a few trades loses.

CPI only against CPI plus other surprises (notebook 05, section 10). Figure A16 and Table A6 compare the CPI-only books (blue) with the books that add the other screened releases (orange and green), against the DTW book from the team’s original slide (gray; ES, 2018–2026, used as a fixed reference in every panel). The trade-off is the same in every equity index. The CPI books earn the most per trade (5–16 bps net) but trade only about 11 times a year. Adding the other releases raises the count to about 95 a year at about 2

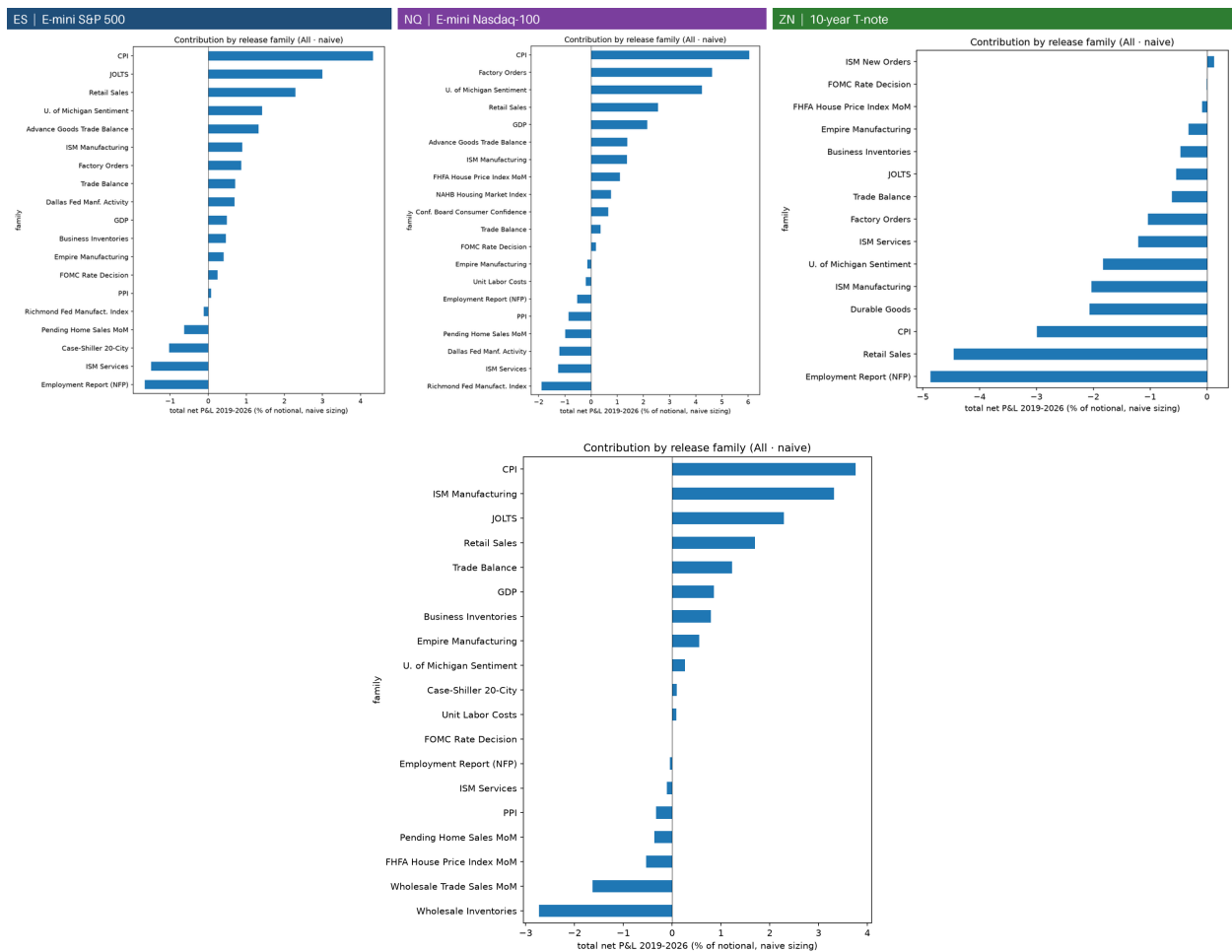


Figure A15. Total net P&L of the broad surprise book by release family, 2019–2026: ES, NQ and ZN (top) and YM (bottom).

bps per trade, and the simple naive sign rule is then the best broad book (0.65 in ES, 0.71 in NQ, 0.68 in YM), close to the CPI books' Sharpe ratio with far more trades, and it keeps the highest Sharpe ratio without 2022 (0.28–0.44). The fitted surprise model and volatility scaling help on CPI but not on the broad book, where scaling mainly adds drawdown (–12 to –14%). Trading a second 30-minute window doubles the trades and removes the edge. In ZN every book loses, and the fitted models almost never trade.

Table A6. CPI only against CPI plus other surprises: net Sharpe ratio (trades per year), 2019–2026, 2 ticks + \$4.50.

Book	ES	NQ	YM	ZN
CPI · naive	0.51 (12)	0.54 (12)	0.54 (11)	−0.67 (12)
CPI · surprise model	0.76 (9)	0.55 (11)	0.45 (9)	−0.37 (0)
CPI · signal × vol	0.77 (12)	0.74 (12)	0.61 (11)	−0.29 (12)
All · naive	0.65 (98)	0.71 (97)	0.68 (93)	−2.42 (82)
All · surprise model	0.11 (43)	0.37 (66)	0.34 (57)	−0.74 (2)
All · signal × vol	0.56 (101)	0.37 (100)	0.29 (96)	−1.11 (84)
Combined (CPI model + others × vol)	0.40 (98)	0.23 (101)	0.16 (95)	−1.25 (72)
All · signal × vol · 2 windows	0.16 (201)	0.00 (199)	−0.12 (193)	−2.02 (168)
All · naive, Sharpe without 2022	0.28	0.44	0.40	−2.53

Notes: CPI books use the average of all CPI lines as the signal, so the naive CPI rule differs from the headline-only rule of notebook 04. “All” is the ten screened families plus CPI.

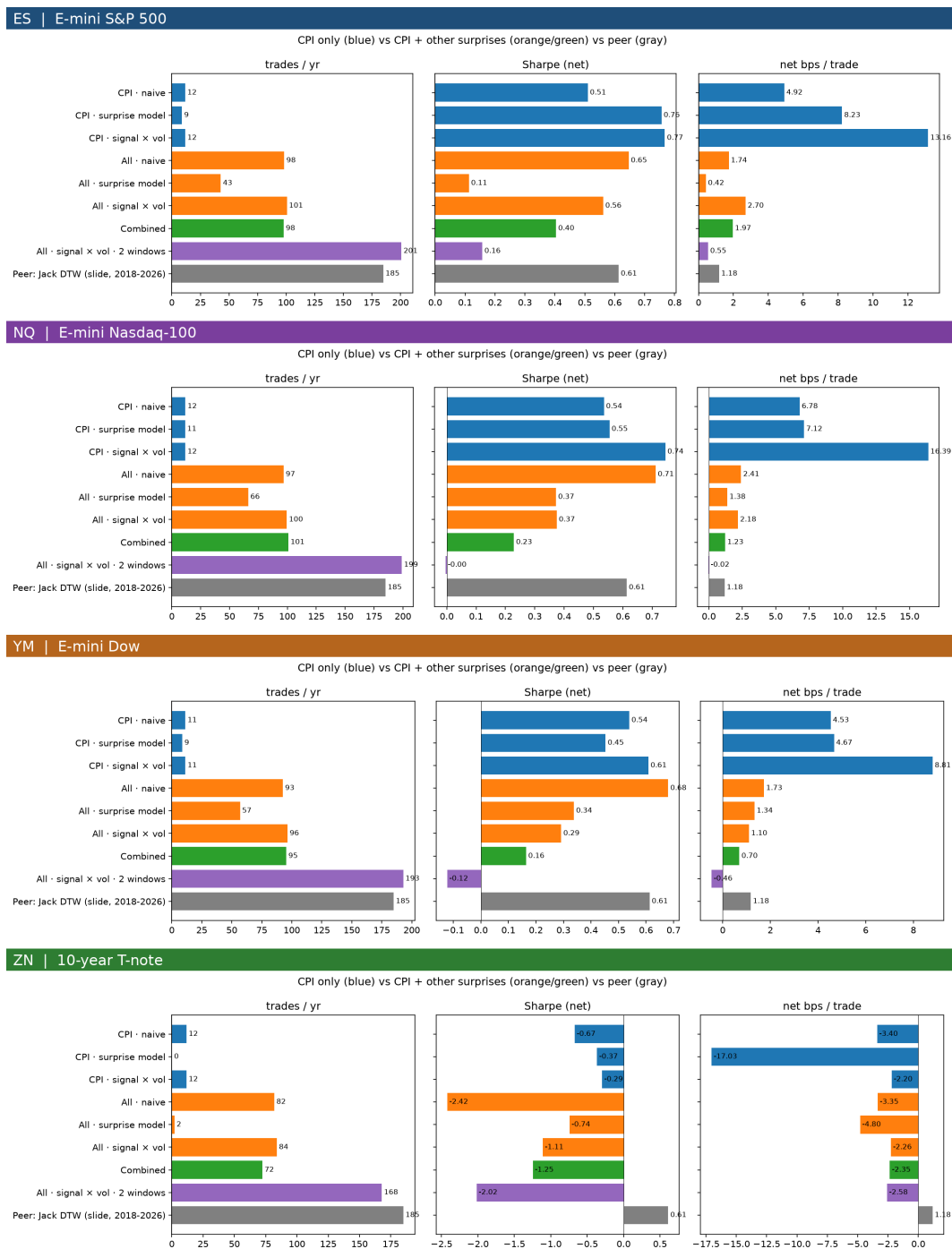


Figure A16. CPI only (blue) against CPI plus other surprises (orange: ten screened families; green: CPI model plus volatility-scaled others; purple: two windows) and the DTW peer reference (gray): trades per year, net Sharpe ratio and net bps per trade, 2019–2026.

E Surprise and DTW (notebook 06)

Cumulative P&L (notebook 06, section 7.3). Figure A17 and Table A7 compare four single-window books on the same releases, walk-forward over 2019–2026 at our base cost: the surprise alone, the DTW direction alone, the surprise kept only when DTW agrees, and a blend of the two signals. On the broad set of releases the surprise alone is the best book in ES (0.65) and NQ (0.71); the DTW-only book loses in ES and is flat in NQ, and requiring agreement removes about half the trades without raising the Sharpe ratio. YM is the exception: there DTW alone earns 0.47 and the blend 1.06, with a third of the drawdown, but the bootstrap probability that the blend is not better than the surprise alone is 6%, and the same combination does not help in ES or NQ (Section V). On CPI alone the four books are close, and most of their cumulative P&L comes from 2022. In ZN every book loses; the surprise book loses most because it trades every release and pays a full tick each time.

Table A7. Surprise, DTW and combined single-window books: net Sharpe ratio, 2019–2026, 2 ticks + \$4.50.

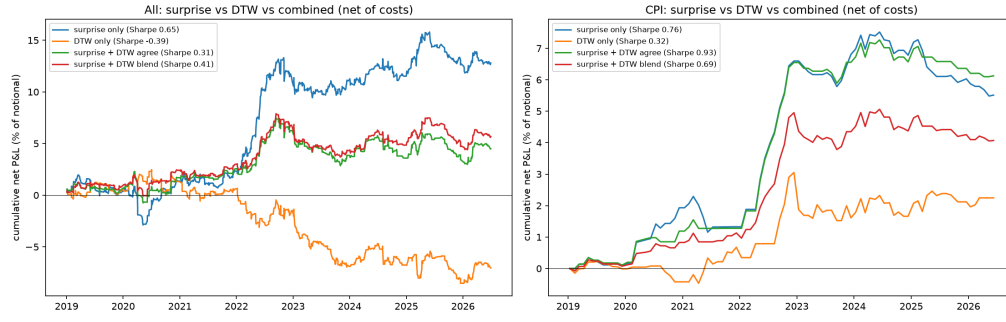
Book	ES	NQ	YM	ZN
<i>All screened releases</i>				
Surprise only	0.65	0.71	0.68	−2.42
DTW only	−0.39	0.08	0.47	−1.25
Surprise + DTW agree	0.31	0.62	0.94	−1.57
Surprise + DTW blend	0.41	0.67	1.06	−1.51
<i>CPI only</i>				
Surprise only	0.76	0.55	0.45	−0.37
DTW only	0.32	0.81	0.41	−0.55
Surprise + DTW agree	0.93	0.54	0.32	—
Surprise + DTW blend	0.69	0.72	0.51	−0.59

“—”: no trades. The ZN CPI surprise book trades once.

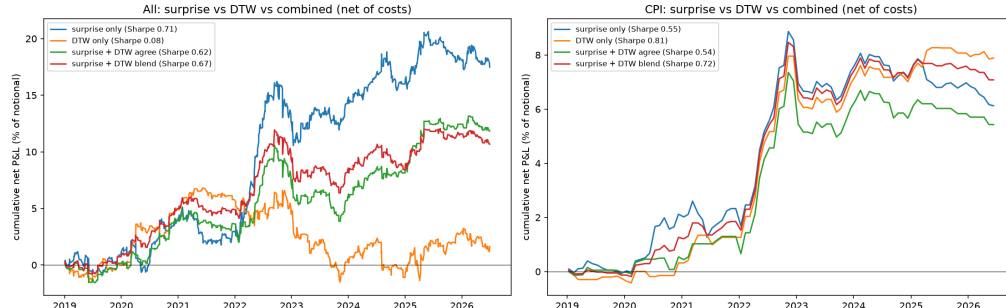
The ladder: can we reach about 185 trades a year? (notebook 06, section 10).

The peer DTW strategy trades about 185 times a year on ES because it takes five consecutive 30-minute trades after each release; our books take one trade per release, which caps them at about 100 trades a year. To answer the supervisor’s question directly—if we trade five windows after each release, how many trades do we get, and what happens to the Sharpe ratio?—notebook 06 copies the ladder structure onto the same releases as the broad surprise book (Table A8). The design was fixed before running. In window k the DTW path is the 31 minutes ending at that window’s entry, matched against earlier releases at the same clock time, and the neighbours’ window- k returns are the forecast; neighbours must have finished

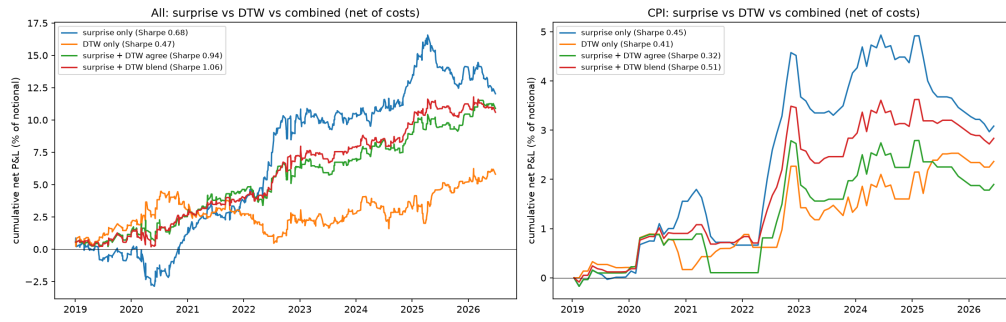
ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



YM | E-mini Dow



ZN | 10-year T-note

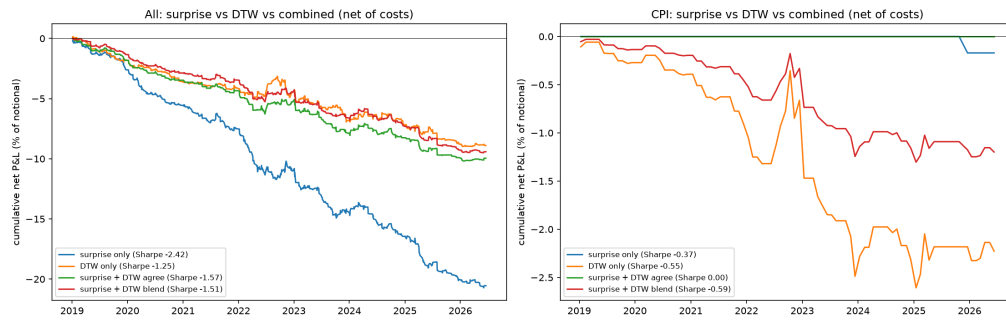


Figure A17. Cumulative net P&L of the surprise-only, DTW-only, surprise + DTW agree and surprise + DTW blend books on all screened releases (left) and on CPI (right), 2019–2026.

window k before the current release, and the gate is computed separately for each window. The surprise rule holds the sign of the release's surprise in each window. Every window pays the same round-trip cost. Three books are compared: a DTW ladder (DTW in all five

windows, closest to the peer), a surprise ladder (the surprise in all five), and the surprise in the first window followed by DTW in windows 2–5. Because the surprise’s effect was already gone in the second window and DTW showed no direction in the first, we expected the trade count to reach the peer’s level while the Sharpe ratio fell toward zero.

Table A8. Ladder design: five back-to-back windows per release.

Window	Entry $\rightarrow$ exit (bars after the release)	Clock time for an 08:30 release
1	close of bar $T \rightarrow T + 29$	08:31 $\rightarrow$ 09:00
2	$T + 29 \rightarrow T + 59$	09:00 $\rightarrow$ 09:30
3	$T + 59 \rightarrow T + 89$	09:30 $\rightarrow$ 10:00
4	$T + 89 \rightarrow T + 119$	10:00 $\rightarrow$ 10:30
5	$T + 119 \rightarrow T + 149$	10:30 $\rightarrow$ 11:00

Later windows of an 08:30 release can overlap the first window of a 10:00 release on the same day; each window is a separate one-contract trade, as in the peer strategy.

The expectation held (Table A9, Figure A18). The ladders reach 300–490 trades a year, well above the peer’s 185, but every one earns less than the single first-window surprise book, and in ES and ZN all of them lose heavily (ES: -1.23 , -1.00 and -0.75). In NQ the surprise in the first window followed by DTW keeps 0.27, and in YM it is flat. The extra trades also lower the hit rate below 50% in ES and ZN, turn the net return per trade negative, and raise the maximum drawdown from about -5% to between -10% and -51% in the equity indices; none of the ladder directions beats random signs at 5%. More trades per release add cost, not information. The peer’s 185 trades a year with a Sharpe ratio of 0.61 come from its own release set and setting; on our releases, copying its structure does not reproduce its result.

Results by window (notebook 06, section 10.3). Table A10 scores each window on its own. The surprise has information only in the first window: its rank correlation with the next 30-minute return is 0.14 in ES, 0.09 in NQ and YM, and then falls to between -0.06 and $+0.05$ in windows 2–5, where every surprise window loses. DTW has no consistent information in any window in ES and NQ (rank correlations of -0.05 to $+0.03$); in YM its first window has a rank correlation of 0.07 and a Sharpe ratio of 0.47, and the later windows fade. In ZN every window loses under both rules. The per-window Sharpe ratios are shown in Figure 10 of the main text.

Table A9. Ladder books: trade count against Sharpe ratio (notebook 06, section 10.4), 2019–2026, 2 ticks + \$4.50.

	Book	Trades/yr	Hit	bps/trade	Total	Sharpe	Ex-2022	Max DD	p (random)
ES	Surprise, window 1 only	98	51%	1.74	12.8%	0.65	0.28	−4.8%	0.00
	DTW ladder	314	46%	−2.06	−48.4%	−1.23	−1.09	−51.2%	0.80
	Surprise ladder	490	46%	−1.15	−42.4%	−1.00	−1.08	−45.6%	0.34
	Surprise W1 + DTW W2–5	348	47%	−1.10	−28.6%	−0.75	−0.80	−30.1%	0.27
NQ	Surprise, window 1 only	97	53%	2.41	17.5%	0.71	0.44	−5.7%	0.00
	DTW ladder	309	50%	−0.13	−2.9%	−0.07	−0.17	−24.2%	0.28
	Surprise ladder	484	50%	−0.39	−14.3%	−0.29	−0.21	−23.6%	0.43
	Surprise W1 + DTW W2–5	342	51%	0.50	12.9%	0.27	−0.04	−15.6%	0.06
YM	Surprise, window 1 only	93	53%	1.73	12.0%	0.68	0.40	−4.5%	0.01
	DTW ladder	298	50%	−0.28	−6.1%	−0.23	−0.14	−12.4%	0.18
	Surprise ladder	463	50%	−0.41	−14.2%	−0.34	−0.44	−17.3%	0.15
	Surprise W1 + DTW W2–5	329	50%	0.00	0.1%	0.00	−0.14	−10.2%	0.06
ZN	Surprise, window 1 only	82	36%	−3.35	−20.5%	−2.42	−2.53	−20.6%	0.81
	DTW ladder	267	32%	−2.70	−54.1%	−3.92	−4.59	−54.2%	0.10
	Surprise ladder	410	30%	−3.13	−96.1%	−7.65	−8.32	−96.1%	1.00
	Surprise W1 + DTW W2–5	294	32%	−2.98	−65.7%	−5.35	−6.27	−65.6%	0.89
	Peer: DTW ladder (slide, ES, 2018–2026)	185	—	1.18	18.6%	0.61	—	−4.9%	—

Net bps per trade; total net return in % of notional; p : share of random-sign books with a Sharpe ratio at least as high.

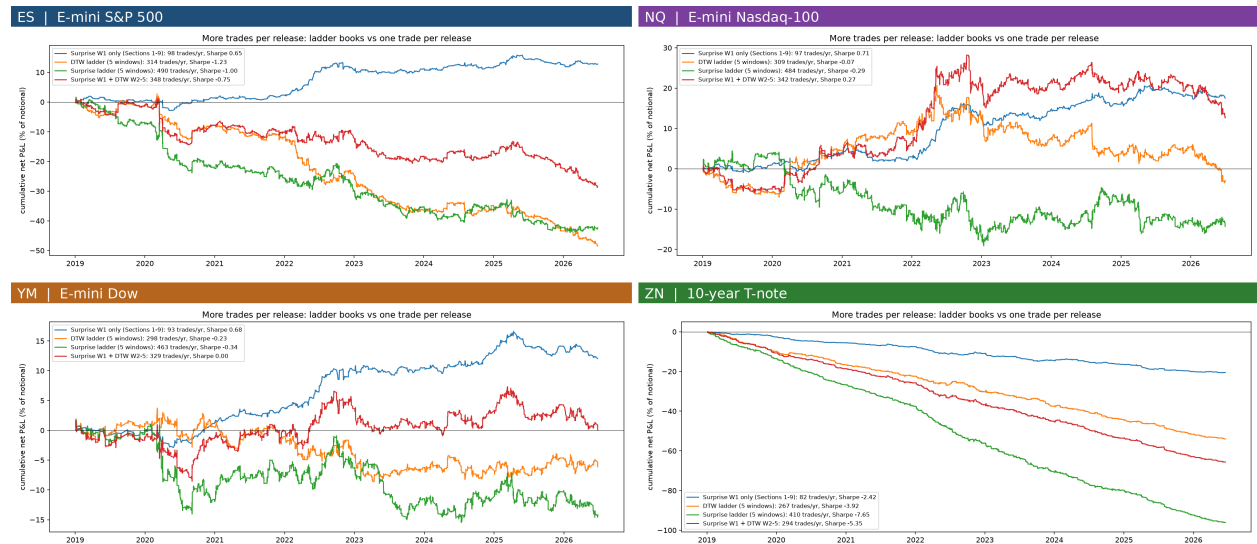


Figure A18. Cumulative net P&L of the five-window ladders (surprise, DTW, and surprise in the first window with DTW afterwards) against the single first-window surprise book, 2019–2026.

Table A10. Each ladder window on its own: rank correlation with the next 30-minute return and net Sharpe ratio, 2019–2026.

Market	Rule	Window 1	Window 2	Window 3	Window 4	Window 5
ES	Surprise	0.14 / 0.65	0.01 / −0.79	−0.06 / −0.57	0.00 / −0.57	0.01 / −1.16
	DTW	0.01 / −0.39	−0.02 / −0.96	0.01 / −0.38	0.03 / −0.22	−0.05 / −1.29
NQ	Surprise	0.09 / 0.71	−0.01 / −0.23	−0.06 / −0.50	0.02 / 0.23	−0.04 / −0.96
	DTW	0.03 / 0.08	−0.01 / 0.01	−0.01 / −0.23	0.02 / 0.15	−0.01 / −0.18
YM	Surprise	0.09 / 0.68	0.05 / −0.15	0.02 / −0.18	−0.06 / −0.80	0.04 / −0.30
	DTW	0.07 / 0.47	0.03 / −0.19	0.05 / 0.15	0.01 / 0.03	−0.06 / −1.03
ZN	Surprise	−0.03 / −2.42	−0.01 / −3.08	0.01 / −3.15	−0.04 / −3.90	0.04 / −3.56
	DTW	0.04 / −1.25	−0.02 / −1.94	−0.02 / −2.34	0.01 / −1.62	−0.01 / −2.92

Each cell: rank correlation / net Sharpe ratio.

F Multi-market surprise book (notebook 07)

Performance by market (notebook 07, section 5). Figure A19 and Table A11 run the same five books, with the same code and constants, on each market. The pattern is the same in the three equity indices and opposite in ZN. The broad naive surprise book earns 0.65 in ES, 0.71 in NQ and 0.68 in YM, with random-sign p -values of 0.006 or below; the CPI model 0.45–0.76; and keeping CPI trades only when DTW agrees 0.32–0.93, on six or seven trades a year. The DTW-only book loses in ES, is flat in NQ and earns 0.47 in YM. Gross against net Sharpe ratios show what costs take: about 0.5 in ES, 0.2 in NQ and 0.3 in YM for the broad book, but 2.1 in ZN, where the naive book goes from -0.30 before costs to -2.42 after them. Over the cost grid the broad book’s Sharpe ratio falls from 1.09 to 0.20 in ES as slippage rises from zero to four ticks, but only from 0.82 to 0.60 in NQ and from 0.91 to 0.45 in YM, because the NQ and YM ticks are smaller relative to their moves.

Table A11. Performance by market, 2019–2026, 2 ticks + \$4.50: net Sharpe ratio (gross in parentheses) and trades per year.

Book	ES	NQ	YM	ZN
All · naive	0.65 (1.17)	0.71 (0.87)	0.68 (1.01)	−2.42 (−0.30)
trades / yr; p (random sign)	98; 0.000	97; 0.005	93; 0.006	82; 0.81
All · DTW only	−0.39 (−0.01)	0.08 (0.21)	0.47 (0.79)	−1.25 (0.47)
All · surprise + DTW agree	0.31 (0.70)	0.62 (0.73)	0.94 (1.21)	−1.57 (−0.12)
CPI · surprise model	0.76 (0.88)	0.55 (0.60)	0.45 (0.53)	−0.37 (1 trade)
CPI · surprise + DTW agree	0.93 (1.02)	0.54 (0.57)	0.32 (0.38)	−
All · naive, Sharpe without 2022	0.28	0.44	0.40	−2.53
<i>All · naive, net Sharpe by round-trip slippage (plus \$4.50)</i>				
0 ticks	1.09	0.82	0.91	−0.56
1 tick	0.87	0.77	0.80	−1.49
2 ticks	0.65	0.71	0.68	−2.42
4 ticks	0.20	0.60	0.45	−4.24

Notes: Monthly Sharpe ratios; “−”: no trades. The ZN CPI model trades about once over the sample, because its predicted drift rarely exceeds the cost.

Portfolios: all markets, equities and ES + NQ (notebook 07, sections 5–6). Table A12 and Figure A20 combine the per-market books with equal-risk weights fixed on 2016–2018 (ES 1.00, NQ 0.71, YM 1.02, ZN 2.67). Adding YM to ES and NQ improves the multi-release equity book: the Sharpe ratio rises from 0.72 to 0.78, total net P&L from 25% to 38%, trades from 195 to 288 a year, and the 90% confidence interval moves fully above zero (0.10 to 1.50). The Sharpe ratio without 2022 also rises (0.38 to 0.43). The cost is a larger drawdown (-9.2% against -7.4%), because the book is bigger. YM helps because its

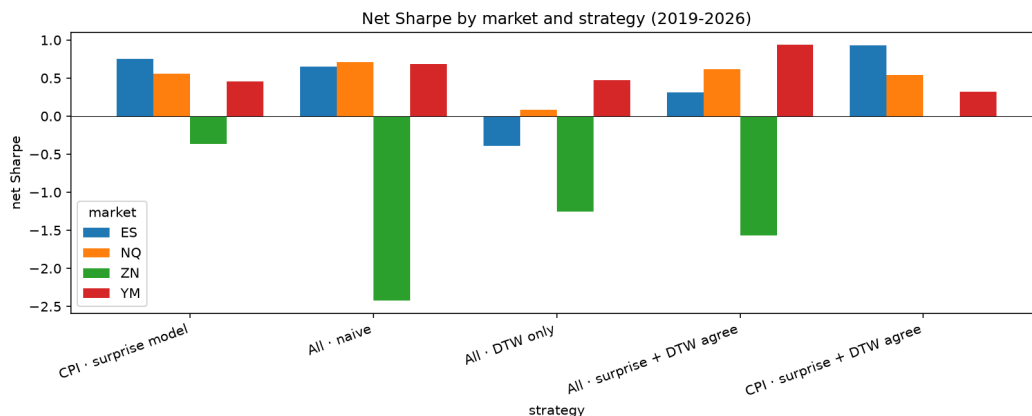


Figure A19. Net Sharpe ratio by market and strategy, 2019–2026, 2 ticks + \$4.50 (notebook 07, section 5).

monthly returns correlate only 0.66 with ES’s and 0.43 with NQ’s. For the CPI-only model YM adds little (0.69 to 0.63), because the CPI books of the three indices are highly correlated. ZN breaks every portfolio it enters: equal-risk weighting gives it the largest weight (2.67, because its moves are small), its multi-release book loses (−2.42), and the four-market naive book is negative (−0.34, drawdown −23%). ZN should therefore be excluded from the traded book. The four-market CPI portfolio still earns 0.61 because the CPI model almost never trades ZN.

Table A12. Equal-risk surprise portfolios across markets, 2019–2026, 2 ticks + \$4.50 (weights fixed on 2016–2018: ES 1.00, NQ 0.71, YM 1.02, ZN 2.67).

Portfolio	Trades/yr	Sharpe	90% CI	Ex-2022	Total	Max DD	p (random)
ES+NQ, all · naive	195	0.72	[−0.01, 1.43]	0.38	25.2%	−7.4%	0.000
Equities (ES+NQ+YM), all · naive	288	0.78	[0.10, 1.50]	0.43	37.5%	−9.2%	0.000
ES+NQ, CPI model	20	0.69	[−0.01, 1.39]	0.02	9.9%	−3.3%	0.023
Equities (ES+NQ+YM), CPI model	29	0.63	[−0.10, 1.37]	−0.04	13.0%	−5.3%	0.020
All four markets, all · naive	370	−0.34	[−1.08, 0.40]	−0.74	−17.4%	−23.4%	0.000
All four markets, CPI model	29	0.61	[−0.13, 1.35]	−0.07	12.6%	−5.8%	0.018
All four markets, DTW only	244	−0.64	[−1.28, −0.01]	−0.62	−23.7%	−29.1%	0.028
All four markets, surprise + DTW agree	195	−0.07	[−0.74, 0.61]	−0.29	−2.6%	−15.6%	0.001
All four markets, CPI + DTW agree	19	0.67	[−0.01, 1.31]	−0.01	11.9%	−3.4%	0.010

Total net return in % of ES-equivalent notional. For the four-market naive book, the random-sign p of 0.000 says its direction is better than random; the loss comes from ZN’s costs, which random-sign books pay too.

Final comparison (notebook 07, section 7). Table A13 brings every market and portfolio book of notebook 07 together. It summarizes the notebook’s conclusion: the consensus surprise is the source of direction in all three equity indices, the broad naive book is the

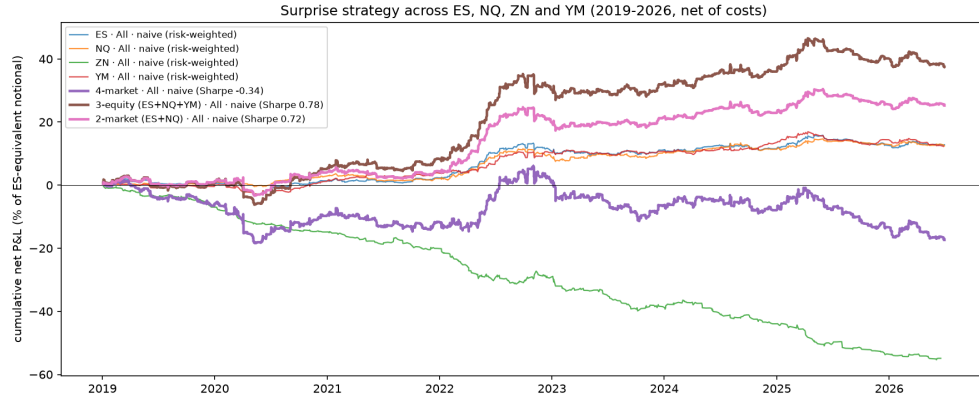


Figure A20. Cumulative net P&L of the risk-weighted surprise books (broad naive book) by market and of the two-, three- and four-market portfolios, 2019–2026.

most robust single book (positive without 2022 and significant against random signs in each index), the three-index portfolio is the best overall, and ZN should not be traded at these costs.

Table A13. Final comparison of the multi-market surprise books, 2019–2026, 2 ticks + \$4.50 (notebook 07, section 7).

Book	Trades/yr	bps/trade	Sharpe	Ex-2022	Max DD	p (random)
ES · CPI · surprise model	9	8.23	0.76	0.05	−2.0%	0.006
ES · All · naive	98	1.74	0.65	0.28	−4.8%	0.000
ES · All · surprise + DTW agree	53	1.14	0.31	0.07	−4.6%	0.030
ES · All · DTW only	65	−1.45	−0.39	−0.32	−11.0%	0.479
NQ · CPI · surprise model	11	7.12	0.55	−0.01	−2.7%	0.047
NQ · All · naive	97	2.41	0.71	0.44	−5.7%	0.005
NQ · All · surprise + DTW agree	51	3.11	0.62	0.59	−6.7%	0.021
NQ · All · DTW only	63	0.34	0.08	0.17	−8.3%	0.271
YM · CPI · surprise model	9	4.67	0.45	−0.15	−2.0%	0.060
YM · All · naive	93	1.73	0.68	0.40	−4.5%	0.006
YM · All · surprise + DTW agree	49	2.94	0.94	0.94	−1.7%	0.002
YM · All · DTW only	61	1.27	0.47	0.58	−4.0%	0.024
ZN · CPI · surprise model	0	−17.03	−0.37	−0.39	−0.2%	1.000
ZN · All · naive	82	−3.35	−2.42	−2.53	−20.6%	0.812
ZN · All · surprise + DTW agree	42	−3.18	−1.57	−1.89	−10.2%	0.645
ZN · All · DTW only	55	−2.15	−1.25	−1.71	−9.1%	0.164
2-market (ES+NQ) · All · naive	195	1.73	0.72	0.38	−7.4%	0.000
2-market (ES+NQ) · CPI · surprise model	20	6.45	0.69	0.02	−3.3%	0.023
3-equity (ES+NQ+YM) · All · naive	288	1.74	0.78	0.43	−9.2%	0.000
3-equity (ES+NQ+YM) · CPI · surprise model	29	5.94	0.63	−0.04	−5.3%	0.020
4-market · CPI · surprise model	29	5.71	0.61	−0.07	−5.8%	0.018
4-market · All · naive	370	−0.63	−0.34	−0.74	−23.4%	0.000
4-market · All · DTW only	244	−1.30	−0.64	−0.62	−29.1%	0.028
4-market · All · surprise + DTW agree	195	−0.18	−0.07	−0.29	−15.6%	0.001
4-market · CPI · surprise + DTW agree	19	8.17	0.67	−0.01	−3.4%	0.010

Net bps per trade; monthly Sharpe ratios; p : share of random-sign books with a Sharpe ratio at least as high.